

Cisco
CCNA

Cisco Certified Network Associate

300 Interview Questions & Answers



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How to use this guide: Each question uses simple, clear language so you can understand concepts easily. Green TIP boxes highlight important exam points. All 10 sections cover the complete CCNA 200-301 syllabus. Prepared by Akhilesh Chaudhari – Senior IT Engineer.

SECTION 01	Network Fundamentals	Q1 – Q30
OSI Model · TCP/IP · IP Addressing · Subnetting · Network Devices · Cables		
Q1	What is a computer network and why do we need it?	
Ans	A computer network connects two or more devices so they can share data, files, printers, and internet access. Without a network, every computer would work alone — you could not send emails, browse websites, or access shared files. Networks make communication fast and efficient.	
TIP	<i>Think of a network like a group of people connected by phone lines — they can talk to each other anytime.</i>	
Q2	What is the OSI model? Explain each layer in simple words.	
Ans	The OSI model is a 7-layer guide that explains how data travels from one computer to another. Layer 1 Physical: actual cables and signals. Layer 2 Data Link: handles MAC addresses and error-free delivery on the same network. Layer 3 Network: handles IP addresses and routing between networks. Layer 4 Transport: TCP and UDP — ensures data arrives completely. Layer 5 Session: opens and closes communication sessions. Layer 6 Presentation: converts data format (encryption, compression). Layer 7 Application: what the user sees — HTTP, DNS, email.	
TIP	<i>Easy trick to remember layers 1 to 7: 'Please Do Not Throw Sausage Pizza Away'</i>	
Q3	What is the difference between TCP and UDP? Give real-life examples.	
Ans	TCP (Transmission Control Protocol) is like sending a registered letter — it confirms delivery, checks for errors, and resends lost data. Used for: websites (HTTP), file downloads (FTP), email. UDP (User Datagram Protocol) is like sending a postcard — fast, no confirmation, no resend. Used for: video calls, online gaming, live streaming. If you miss a few packets in a live video, it is acceptable. If you miss data in a file download, it is a problem.	
TIP	<i>TCP = reliable but slower. UDP = fast but no guarantee. CCNA exam often asks when to use each.</i>	
Q4	What is the TCP 3-way handshake?	
Ans	Before two devices communicate via TCP, they do a 3-step process: Step 1 — Client sends SYN (I want to connect). Step 2 — Server replies SYN-ACK (OK, I am ready). Step 3 — Client sends ACK (Great, let us start). This ensures both sides are ready before any data is sent. Think of it like: 'Hello?' □ 'Hello, I hear you!' □ 'Great, let us talk.'	

Q5 What is an IP address? What is the difference between IPv4 and IPv6?

Ans An IP address is a unique number given to every device on a network — like a home address for computers. IPv4 uses 32 bits written as 4 numbers: 192.168.1.1. It supports about 4.3 billion addresses — which are running out. IPv6 uses 128 bits written in hexadecimal: 2001:db8::1. It supports trillions of addresses. The world is slowly moving from IPv4 to IPv6 because IPv4 addresses are exhausted.

Q6 What are private IP address ranges and why are they used?

Ans Private IPs are used inside your home or office network — they are NOT visible on the internet. Three ranges: 10.0.0.0 – 10.255.255.255 (large organisations), 172.16.0.0 – 172.31.255.255 (medium networks), 192.168.0.0 – 192.168.255.255 (home and small office — most common). Your home router gives your devices private IPs, and uses ONE public IP to represent all of them on the internet via NAT.

Q7 What is subnetting? Explain with a simple example.

Ans Subnetting divides one large network into smaller sections called subnets. Why? To reduce traffic, improve security, and organise devices. Example: 192.168.1.0/24 gives you 254 usable addresses. If you subnet to /25, you get two groups of 126 addresses each — one for staff PCs, one for printers. Formula: Usable hosts = $2^{(\text{host bits})} - 2$. /24 = 8 host bits = 254 hosts.

TIP *Subnetting is one of the most important topics in CCNA — practice it daily!*

Q8 What is a subnet mask? What does /24 mean?

Ans A subnet mask tells the device which part of an IP address is the network and which part is the host. /24 means the first 24 bits are the network portion = 255.255.255.0. So in 192.168.1.50/24, the network is 192.168.1.0 and the host is .50. Devices with the same network portion can talk directly. Devices on different networks need a router.

Q9 What is a default gateway?

Ans A default gateway is the IP address of the router on your local network. When your computer wants to send data to a device outside your network (like accessing Google), it sends the data to the default gateway first. The router then forwards it towards the destination. Without a default gateway, your device can only communicate with devices on the same local network.

Q10 What is DHCP and how does it work?

Ans DHCP (Dynamic Host Configuration Protocol) automatically assigns IP addresses to devices. Without it, you would manually type an IP address on every computer. How it works (DORA): Discover — device broadcasts 'I need an IP address'. Offer — DHCP server replies with an available IP. Request — device says 'I want that IP'. Acknowledge — server confirms. Your home router runs a DHCP server by default.

TIP DORA = Discover, Offer, Request, Acknowledge — memorise this for CCNA!

Q11 What is DNS and why is it important?

Ans DNS (Domain Name System) is like the internet's phone book. Instead of remembering that Google is at 142.250.182.46, you just type google.com — DNS converts that name into the IP address automatically. Without DNS, you would have to type IP addresses to visit every website. DNS uses port 53. When you type a website, your computer first asks a DNS server for the IP address, then connects to that IP.

Q12 What is NAT (Network Address Translation)?

Ans NAT allows many devices with private IP addresses to share a single public IP address when accessing the internet. Your home router does this. Example: Your laptop has private IP 192.168.1.5. When you visit a website, the router replaces your private IP with its public IP (say 203.0.113.10) in the packet. The website sees only 203.0.113.10. When the reply comes back, the router forwards it to your laptop. NAT conserves public IPs and adds a layer of security.

Q13 What is the difference between a hub, switch, and router?

Ans Hub: sends data to ALL devices connected — wasteful and creates collisions. Old technology, rarely used now. Switch: smart hub — remembers which device is on which port (MAC address table) and sends data ONLY to the correct device. Efficient, used in all modern networks. Router: connects different networks together and routes traffic between them using IP addresses. Your home device combining all three is technically a router with a built-in switch and WAP.

Q14 What is a MAC address?

Ans A MAC (Media Access Control) address is a unique identifier burned into every network card during manufacturing. It looks like: 00:1A:2B:3C:4D:5E — 12 hexadecimal characters. Unlike IP addresses (which can change), a MAC address is permanent. Switches use MAC addresses to decide where to send frames within the same network. You can see your MAC address with 'ipconfig /all' (Windows) or 'ip addr' (Linux).

Q15 What is ARP (Address Resolution Protocol)?

Ans ARP finds the MAC address of a device when you only know its IP address. Example: Your PC knows the server's IP is 192.168.1.10 but needs its MAC address to send data at Layer 2. ARP broadcasts: 'Who has 192.168.1.10?' The server replies with its MAC address. Your PC saves this in its ARP cache to avoid asking again. Type 'arp -a' in your command prompt to see cached ARP entries.

Q16 What are the different types of network topologies?

Ans Bus: all devices on one cable — if cable breaks, whole network fails (obsolete). Ring: devices connected in a circle — data travels one direction (obsolete). Star: all devices connect to a central switch — most common today. If one cable fails, only that device is affected. Mesh: every device connects to every other — expensive but extremely reliable (used in data centres and WAN). Hybrid: combination of star and mesh — used in large enterprises.

Q17 What is the difference between a LAN, WAN, and MAN?

Ans LAN (Local Area Network): small area — home, office, single building. Fast (1 Gbps+), low cost, privately owned. WAN (Wide Area Network): large geographic area — connects cities, countries. The internet is the world's largest WAN. Slower, more expensive, often leased from providers. MAN (Metropolitan Area Network): covers a city — like a city-wide internet service. Larger than LAN but smaller than WAN. Used by city governments, universities.

Q18 What is a VLAN and why is it used?

Ans VLAN (Virtual LAN) logically separates devices on the same physical switch into different groups, as if they were on separate networks. Example: Separate HR computers from Engineering computers on the same switch — HR cannot access Engineering files even though they share the same physical switch. Benefits: better security, less broadcast traffic, easier management. VLANs are configured on managed switches.

TIP *VLANs are a core CCNA topic — understand trunking and inter-VLAN routing too.*

Q19 What is a broadcast domain?

Ans A broadcast domain is a group of devices that all receive the same broadcast message. In a simple network, if one device sends a broadcast, ALL devices on that network receive it. This creates traffic. Routers separate broadcast domains — broadcasts do not cross routers. VLANs also create separate broadcast domains on a switch. Smaller broadcast domains = better network performance.

Q20 What is the difference between full-duplex and half-duplex?

Ans Half-duplex: data flows in one direction at a time — like a walkie-talkie. When one person talks, the other must wait. Increases collision risk. Full-duplex: data flows in both directions simultaneously — like a phone call. Both sides send and receive at the same time. Modern switches support full-duplex. Mismatch between a switch port and a device (one full, one half) causes poor performance.

Q21 What are the main cable types used in networking?

Ans UTP (Unshielded Twisted Pair): most common, copper, used for Ethernet. Categories: Cat5e (1 Gbps), Cat6 (10 Gbps up to 55m), Cat6a (10 Gbps up to 100m). STP (Shielded): has metal shield to reduce interference — used in factories. Coaxial: older cable, used for cable TV/internet. Fibre optic: uses light, not electricity — extremely fast and long distance. Two types: Single-mode (long distance, yellow), Multi-mode (shorter distance, orange/aqua).

Q22 What is the difference between single-mode and multi-mode fibre?

Ans Single-mode fibre (SMF): very thin core (9 microns), carries one beam of light, travels very long distances (up to 100 km), uses laser light, more expensive, yellow jacket. Used between cities and for long campus runs. Multi-mode fibre (MMF): thicker core (50-62.5 microns), multiple light paths, shorter distance (up to 550m), less expensive, orange or aqua jacket. Used inside buildings and data centres.

Q23 What is a straight-through vs a crossover cable?

Ans Straight-through cable: both ends wired identically — used to connect different types of devices (PC to switch, router to switch). Most common type. Crossover cable: one end wired differently — used to connect same types of devices (switch to switch, PC to PC directly). Modern switches auto-detect (auto-MDI/MDIX) and can use either cable type, making crossover cables largely unnecessary today.

Q24 What is a collision domain?

Ans A collision domain is a network segment where two devices transmitting simultaneously cause a collision (data crash). Hubs create one large collision domain — all devices share it. Switches eliminate collisions by giving each port its own collision domain. So a 24-port switch has 24 collision domains. This is why switches replaced hubs — no more collisions = better performance.

Q25 What is CIDR notation?

Ans CIDR (Classless Inter-Domain Routing) notation writes the subnet mask as a slash and number after the IP address. Example: 192.168.1.0/24 means the first 24 bits are the network. Quick reference: /24 = 254 hosts (255.255.255.0), /25 = 126 hosts, /26 = 62 hosts, /27 = 30 hosts, /28 = 14 hosts, /29 = 6 hosts, /30 = 2 hosts (used for router-to-router links).

Q26 What is the loopback address?

Ans The loopback address 127.0.0.1 (IPv4) or ::1 (IPv6) allows a device to send traffic to itself for testing. When you 'ping 127.0.0.1', the packet never leaves your computer — it tests whether your own TCP/IP stack is working correctly. If this fails, your network software is broken. It is like writing a letter to your own address to test the postal system.

Q27 What is APIPA? When does a device get a 169.254.x.x address?

Ans APIPA (Automatic Private IP Addressing) kicks in when a device cannot find a DHCP server. The device assigns itself an address in the range 169.254.0.0 – 169.254.255.255. This allows communication with other APIPA devices on the same network, but NOT internet access. Seeing 169.254.x.x is a warning sign — it means DHCP is not working. Check your DHCP server or cable connection.

TIP

169.254.x.x = DHCP failure. This diagnosis appears directly on CCNA exams!

Q28 What is the difference between a managed and unmanaged switch?

Ans Unmanaged switch: plug and play, no configuration needed, basic connectivity, no VLANs, no monitoring, cheap. Good for home use. Managed switch: fully configurable via CLI or web GUI. Supports VLANs, STP, port security, SNMP monitoring, QoS, link aggregation, and more. Used in all enterprise networks. Cisco switches are always managed. You configure them using Cisco IOS commands.

Q29 What is a network protocol?

Ans A network protocol is a set of rules that devices follow to communicate. Just like humans need a common language to talk, computers need protocols. Examples: HTTP — for web pages. SMTP — for sending email. FTP — for file transfer. DNS — for name resolution. TCP/IP — for internet communication. Without protocols, devices from different manufacturers could not communicate with each other.

Q30 What are well-known port numbers you must know for CCNA?

Ans Critical port numbers: FTP 20/21, SSH 22, Telnet 23, SMTP 25, DNS 53, DHCP 67/68, HTTP 80, POP3 110, NTP 123, IMAP 143, HTTPS 443, SMB 445, RDP 3389, SNMP 161/162, RADIUS 1812/1813, TACACS+ 49, Syslog 514. Ports 0-1023 are well-known ports. Ports 1024-49151 are registered ports. Ports 49152-65535 are dynamic/private ports.

TIP *Port numbers appear on almost every CCNA exam — memorise at least the top 15!*

SECTION 02

Cisco IOS & Device Configuration

Q31 – Q60

Cisco CLI · IOS Modes · Basic Config · Passwords · SSH · Router/Switch Setup

Q31 What is Cisco IOS?

Ans Cisco IOS (Internetwork Operating System) is the operating system that runs on Cisco routers and switches. It is like Windows for Cisco devices. You configure Cisco devices using text commands typed in a terminal — this is called the Command Line Interface (CLI). You access the CLI by connecting a console cable from your laptop to the device's console port and using a terminal program like PuTTY.

TIP *Cisco IOS commands are the heart of the CCNA exam. Practice them in Cisco Packet Tracer daily!*

Q32 What are the IOS configuration modes? How do you move between them?

Ans Three main modes: User EXEC mode (Router>): limited view-only access. Privileged EXEC mode (Router#): full access to view all settings. Type 'enable' to enter. Global Configuration mode (Router(config)#): change settings. Type 'configure terminal' or 'conf t'. Sub-modes: Interface config (Router(config-if)#): configure a specific port. Line config (Router(config-line)#): configure console or VTY. To go back, use 'exit' or press Ctrl+Z to jump to privileged mode.

Q33 What is the difference between 'show running-config' and 'show startup-config'?

Ans Running config (RAM): the configuration currently active on the device. Changes take effect immediately but are LOST if the device reboots. Startup config (NVRAM): the configuration that loads when the device starts. To save your changes permanently, type 'copy running-config startup-config' (or 'wr'). If you do not save and the router reboots, all your changes are gone. Always save after making changes!

TIP *Forgetting to save config is a common mistake. Always run 'copy run start' after changes!*

Q34 How do you configure a hostname on a Cisco device?

Ans In Global Configuration mode, type: hostname Router1. The prompt changes immediately: Router(config)# becomes Router1(config)#. Hostnames help identify devices in large networks. Best practice: use a descriptive name like 'CoreSwitch-Floor3' or 'Branch-Router-Mumbai'. This makes network diagrams and troubleshooting much easier.

Q35 How do you set passwords on a Cisco router?

Ans Three important passwords: Enable password (weak, avoid): enable password cisco123. Enable secret (strong, encrypted, preferred): enable secret cisco123. Console password: line console 0 □ password cisco □ login. VTY (Telnet/SSH) password: line vty 0 4 □ password cisco □ login. Always use 'enable secret' over 'enable password' because secret is encrypted with MD5 in the config file.

Q36 What is the 'service password-encryption' command?

Ans This command encrypts all plaintext passwords stored in the configuration file. Without it, passwords are visible as plain text in 'show running-config' — anyone who views the config can steal them. With service password-encryption: type 7 encryption is applied (weak but better than nothing). Note: 'enable secret' uses MD5 and is already encrypted regardless of this command. Use both for best security.

Q37 How do you configure an IP address on a router interface?

Ans Go to the interface: interface gigabitethernet 0/0 (or 'int g0/0'). Set IP: ip address 192.168.1.1 255.255.255.0. Enable the interface: no shutdown. By default, all router interfaces are shut down. Verify: show interfaces or show ip interface brief. The 'no shutdown' command is critical — without it, the interface will not work even with a correct IP address configured.

Q38 What is 'no shutdown' and why is it needed?

Ans On Cisco routers, all interfaces are administratively disabled by default (status shows 'administratively down'). The 'no shutdown' command enables the interface. On Cisco switches, ports are enabled by default. To disable a port: use 'shutdown'. Always run 'show ip interface brief' after configuring interfaces to verify they are up/up (first = line protocol, second = layer 2).

Q39 How do you configure a Cisco switch port for a specific VLAN?

Ans First create the VLAN: vlan 10 □ name Sales. Then assign port to VLAN: interface fastethernet 0/1 □ switchport mode access □ switchport access vlan 10. The port is now an access port in VLAN 10. Devices connected here are in the Sales VLAN and can only communicate with other Sales VLAN devices unless a router or Layer 3 switch routes between VLANs.

Q40 What is a trunk port on a switch?

Ans A trunk port carries traffic for multiple VLANs on a single cable — typically connecting switch to switch, or switch to router. Uses 802.1Q tagging to label which VLAN each frame belongs to. Configure: interface g0/1 `switchport mode trunk`. You can specify which VLANs are allowed: `switchport trunk allowed vlan 10,20,30`. Native VLAN (default VLAN 1) traffic is sent without a tag.

Q41 What is 802.1Q trunking?

Ans 802.1Q is the industry standard for VLAN tagging on trunk links. When a frame enters a trunk port, a 4-byte tag is inserted into the Ethernet frame containing the VLAN ID (1-4094). The receiving switch reads the tag and forwards the frame to the correct VLAN. Without this tag, switches would not know which VLAN a frame belongs to when it crosses a trunk link. Cisco switches use 802.1Q by default.

Q42 What is the native VLAN?

Ans The native VLAN is the VLAN whose traffic is sent across a trunk link WITHOUT a tag. By default, this is VLAN 1. Any untagged traffic received on a trunk port is placed into the native VLAN. Both ends of a trunk link must have the same native VLAN — mismatch causes connectivity problems. Security best practice: change native VLAN from 1 to an unused VLAN to prevent VLAN hopping attacks.

Q43 What is inter-VLAN routing?

Ans Devices in different VLANs cannot communicate by default — a router is needed. Two methods: Router-on-a-Stick (ROAS): one physical router port connected to a trunk port, with sub-interfaces for each VLAN (e.g. `int g0/0.10`, `encapsulation dot1Q 10`, `ip address 192.168.10.1`). Layer 3 Switch: configures SVIs (virtual interfaces) for each VLAN directly on the switch — faster than ROAS. Most enterprise networks use Layer 3 switches.

Q44 How do you configure SSH on a Cisco device?

Ans SSH is secure remote access. Steps: Set hostname (required). Set domain name: `ip domain-name company.com`. Generate RSA keys: `crypto key generate rsa` (use 2048 bits). Create local user: `username admin secret cisco123`. Configure VTY lines: `line vty 0 4` `transport input ssh` `login local`. Enable SSH version 2: `ip ssh version 2`. Verify: `show ip ssh`. Never use Telnet in production — it sends passwords in plain text.

TIP

Always prefer SSH over Telnet. This is tested in CCNA security topics!

Q45 What is the 'show' command and its most useful variants?

Ans Show commands display information without changing anything — safe to run anytime. Key show commands: show running-config (full current config), show ip interface brief (all interfaces — IP and status), show interfaces (detailed stats), show version (IOS version, uptime, memory), show ip route (routing table), show vlan brief (VLAN list), show mac address-table (CAM table), show cdp neighbors (connected Cisco devices).

Q46 What is CDP (Cisco Discovery Protocol)?

Ans CDP is Cisco's proprietary protocol that automatically discovers neighbouring Cisco devices. It runs on Layer 2 and shares: device type, IOS version, IP address, connected port, and capabilities. Command: show cdp neighbors detail. Very useful for mapping a network when documentation is missing. Security note: disable CDP on external-facing ports (no cdp enable on interface) — it reveals too much info to potential attackers.

Q47 What is LLDP?

Ans LLDP (Link Layer Discovery Protocol) is the open-standard version of CDP, supported by all vendors (not just Cisco). Cisco devices run CDP by default but can also run LLDP. Enable LLDP globally: lldp run. Enable on interface: lldp transmit and lldp receive. View neighbours: show lldp neighbors. Use LLDP when working in multi-vendor environments (mix of Cisco, HP, Juniper devices).

Q48 What is the Cisco IOS boot process?

Ans When a Cisco router powers on: POST (Power-On Self Test) — checks hardware. Bootstrap — small program finds and loads IOS. IOS loads from Flash memory (default). Configuration loads from NVRAM (startup-config). If no startup-config, enters setup mode or blank config. The 'show version' command shows where IOS was loaded from and the config register setting (normally 0x2102 = normal boot).

Q49 What is the configuration register?

Ans The configuration register is a 16-bit value telling the router how to boot. Common values: 0x2102 — normal boot (loads IOS from flash, config from NVRAM). 0x2142 — password recovery mode (ignores startup-config on boot — used to reset forgotten passwords). 0x2100 — boot to ROM Monitor (ROMMON) mode. Check with 'show version' (last line). Change with 'config-register 0x2102' in global config.

Q50 How do you recover a forgotten enable password on a Cisco router?

Ans Password recovery process: Power off router. Power on while holding Ctrl+Break (or Break key) to enter ROMMON mode. Type 'confreg 0x2142' (ignore startup-config on next boot). Type 'reset' to reboot. Router boots without loading saved config. Enter 'enable' (no password needed). Copy startup-config to running-config: 'copy startup-config running-config'. Change the password. Save config. Reset register: 'config-register 0x2102'. Save again and reload.

Q51 What is 'no ip domain-lookup' and why is it used?

Ans By default, if you type an unrecognised command in Cisco IOS, the router tries to resolve it as a hostname using DNS — this causes a long delay (looks like it is hanging). 'no ip domain-lookup' disables this. It is the first command most Cisco engineers type in a lab or exam simulator. In production, only use it if you have a specific reason — otherwise DNS lookups may be needed for the device.

Q52 What is banner motd on Cisco?

Ans Banner MOTD (Message of the Day) displays a warning message to anyone who connects to the device. It should say something like: 'Authorised access only. Unauthorised access is prohibited.' This is legally important — courts have ruled that without a warning banner, accessing a device may not be prosecutable as unauthorised. Configure: banner motd # Authorised users only! #

Q53 What is the difference between console, VTY, and AUX lines?

Ans Console line (line con 0): direct physical connection via console cable. Used for initial setup and password recovery. VTY lines (line vty 0 4 or 0 15): virtual terminal lines for remote access via Telnet or SSH. '0 4' allows 5 simultaneous remote sessions. AUX line (line aux 0): auxiliary port for dial-up modem access — very rarely used today. Configure passwords and login on each line type separately.

Q54 What is Cisco Packet Tracer?

Ans Cisco Packet Tracer is a free network simulation tool by Cisco. It lets you build virtual networks with simulated routers, switches, PCs, servers, and wireless devices — without needing real hardware. You can practice all CCNA commands and scenarios in it. Download it free from Cisco's NetAcad website. It is the number one tool for CCNA exam preparation. GNS3 is another option that runs real IOS images.

Q55 What is the 'ping' command and what does it test?

Ans Ping sends ICMP Echo Request packets to a destination and waits for replies. It tests basic IP connectivity between two devices. Output: !!!!! = 5 out of 5 packets received (success). = 0 out of 5 (no connectivity). On Cisco IOS, '!!! ..' means partial success (e.g. first two failed due to ARP resolution). Extended ping (from Router#, type 'ping' and press Enter) gives more options.

Q56 What is the 'traceroute' command?

Ans Traceroute shows the path packets take from your device to a destination, listing every router (hop) along the way with its IP and response time. Use it to find exactly WHERE a network problem is occurring. Example: if hop 3 shows * * * (no reply), there is a problem at or after hop 3. On Windows: 'tracert'. On Linux and Cisco: 'traceroute'. Very powerful troubleshooting tool.

Q57 What is 'show ip interface brief' and how to interpret it?

Ans This is the most-used command on Cisco routers. It shows all interfaces, their IP addresses, and status. Status has two parts: Line status (Physical layer — is cable connected?) and Protocol status (Data Link layer — is communication happening?). Up/Up = working. Up/Down = Layer 1 OK but Layer 2 problem. Down/Down = no cable or 'shutdown' applied. Administratively down/down = 'shutdown' was manually applied.

Q58 What is the 'write' or 'copy run start' command?

Ans Both commands save the current running configuration to NVRAM (startup config) so it survives a reboot. 'copy running-config startup-config' is the full command. 'wr' is the shortcut. Run this after EVERY configuration change. If you forget and the device reboots (power cut, crash), all your changes are lost. Think of it like pressing Save in a document editor.

Q59 How do you erase all configuration from a Cisco device?

Ans To completely reset a Cisco router: write erase (deletes startup-config from NVRAM) □ reload (restarts the device) □ say No when asked to save □ device boots with factory defaults. For a switch, also type 'delete flash:vlan.dat' before reloading to remove VLAN database. This is useful in a lab when you want to start fresh or return a device to its original state.

Q60 What is clock rate on a Cisco serial interface?

Ans On a serial WAN link between two Cisco routers (in a lab), one end is the DCE (Data Communications Equipment) and the other is the DTE. The DCE end must provide clocking. Command: `clock rate 64000` (sets 64 Kbps). Check which end is DCE: `show controllers serial 0/0` (DCE = must configure clock rate). If clock rate is missing on DCE end, the link will not come up. In real WAN connections, the service provider provides clocking.

SECTION 03

Routing Concepts & Protocols

Q61 – Q90

Static Routes · RIP · OSPF · EIGRP · BGP · Routing Table · Default Routes

Q61 What is routing and how does a router make forwarding decisions?

Ans Routing is the process of choosing the best path for data to travel from source to destination across multiple networks. A router looks at the destination IP in each packet, searches its routing table for the best matching route, and forwards the packet out the correct interface. The routing table contains: directly connected networks, static routes manually configured, and dynamically learned routes from routing protocols.

TIP *The routing table is the router's map. Understanding it is essential for all CCNA routing topics.*

Q62 What is a routing table? How do you view it?

Ans The routing table lists all networks a router knows how to reach. View it with: `show ip route`. Reading the output: C = Connected (directly attached network), S = Static route (manually configured), R = RIP learned, O = OSPF learned, D = EIGRP learned, * = default route. Each entry shows: network/prefix, next-hop IP or exit interface, and metric. The router picks the most specific (longest prefix) matching route.

Q63 What is a static route? How do you configure one?

Ans A static route is a route you manually type into the router — it does not change automatically. Syntax: `ip route [destination network] [subnet mask] [next-hop IP or exit interface]`. Example: `ip route 192.168.2.0 255.255.255.0 10.0.0.2` — tells the router that to reach 192.168.2.0/24, send to next-hop 10.0.0.2. Use static routes for small networks, specific paths, or backup routes.

Q64 What is a default route?

Ans A default route (0.0.0.0/0) is the route used for ANY destination not specifically listed in the routing table. It is the 'catch-all' route. Configure: `ip route 0.0.0.0 0.0.0.0 [ISP-gateway-IP]`. All internet-bound traffic is forwarded to the ISP. In routing table shown as S* (static default) or it can be learned dynamically. Every internet-connected router has a default route pointing toward the internet.

Q65 What is the difference between static and dynamic routing?

Ans Static routing: you manually type every route. Simple, predictable, no protocol overhead. But if a link fails, the route stays in the table — no automatic failover. Good for small networks with few routes. Dynamic routing: routers share route information automatically using protocols (OSPF, EIGRP, RIP). Automatically updates when topology changes. More complex to configure but scales well and self-heals when links fail.

Q66 What is Administrative Distance (AD)?

Ans When a router learns about the same network from multiple sources, AD decides which one to trust most. Lower AD = more trusted = preferred. AD values: Connected = 0 (most trusted), Static = 1, EIGRP = 90, OSPF = 110, RIP = 120, External EIGRP = 170. Example: if both OSPF and RIP tell the router about 192.168.1.0/24, the router uses the OSPF route because OSPF has lower AD (110 < 120).

TIP *AD decides WHICH routing protocol's route to use. Metric decides the BEST path within a protocol.*

Q67 What is a routing metric?

Ans A metric is the value a routing protocol uses to choose the best path when multiple paths to the same destination exist. Different protocols use different metrics: RIP uses hop count (fewer hops = better). OSPF uses cost (based on bandwidth — higher bandwidth = lower cost = better). EIGRP uses composite metric (bandwidth + delay). Lower metric = preferred path. Metric only compares routes from the SAME protocol.

Q68 What is RIP (Routing Information Protocol)?

Ans RIP is a simple, old dynamic routing protocol. It uses hop count as metric — maximum 15 hops (16 = unreachable). It sends its full routing table to neighbours every 30 seconds. Very easy to configure but slow to converge and not scalable. RIPv1 is classful (no subnet mask sent), RIPv2 is classless (sends subnet mask). RIPv2 also supports authentication. Configure: `router rip` □ `version 2` □ `network 192.168.1.0`.

Q69 What is OSPF (Open Shortest Path First)?

Ans OSPF is a popular, fast, and scalable routing protocol used in enterprise networks. It uses 'cost' as its metric ($\text{cost} = 100 \text{ Mbps} / \text{interface bandwidth}$). It builds a complete map of the network (link-state database) and runs Dijkstra's shortest path algorithm. Features: fast convergence (typically seconds), classless, supports large networks, uses areas (Area 0 is the backbone). OSPF is the most important routing protocol in CCNA.

TIP *OSPF is heavily tested in CCNA. Know DR/BDR election, areas, and LSA types!*

Q70 What is OSPF DR and BDR election?

Ans On Ethernet networks with multiple OSPF routers, all routers form neighbour relationships with just two special routers: DR (Designated Router) and BDR (Backup DR). This reduces OSPF traffic. Election process: highest OSPF priority wins (default 1). Tie: highest Router ID wins. Router ID: manually set, or highest loopback IP, or highest interface IP. Configure priority: `ip ospf priority 100` (on the interface). Priority 0 = never become DR.

Q71 What is EIGRP?

Ans EIGRP (Enhanced Interior Gateway Routing Protocol) is Cisco's advanced routing protocol. It is faster than OSPF for convergence, easy to configure, and supports unequal-cost load balancing (send traffic across multiple paths with different speeds). Uses composite metric (bandwidth + delay by default). EIGRP uses DUAL algorithm to calculate routes and keeps backup routes (feasible successors) ready for instant failover. Now an open standard (RFC 7868).

Q72 What is OSPF cost and how is it calculated?

Ans $\text{OSPF cost} = \text{Reference Bandwidth} / \text{Interface Bandwidth}$. Default reference bandwidth = 100 Mbps. So: 100 Mbps link = cost 1, 10 Mbps = cost 10, 1 Mbps = cost 100. Problem: modern networks have 1 Gbps links — they get the same cost as 100 Mbps! Fix: change reference bandwidth: `auto-cost reference-bandwidth 1000` (for gigabit networks). Always set this consistently on ALL OSPF routers in your network.

Q73 What is BGP (Border Gateway Protocol)?

Ans BGP is the routing protocol of the internet — it routes traffic between different organisations (Autonomous Systems). It is extremely complex and not usually on the CCNA level, but you should know the basics. eBGP: between different organisations. iBGP: within the same organisation. BGP uses TCP port 179. It bases routing on policies and AS path (list of networks the route has passed through) rather than just bandwidth.

Q74 What is a loopback interface?

Ans A loopback interface is a virtual interface on a router — it always stays up as long as the router is on. It is not connected to any physical cable. Uses: OSPF Router ID (use loopback IP for stability), management access (always reachable), testing. Configure: `interface loopback 0` □ `ip address 1.1.1.1 255.255.255.255`. Typically use /32 mask (single host address). Many engineers assign loopbacks like 1.1.1.1/32 for Router 1, 2.2.2.2/32 for Router 2.

Q75 What is the difference between a routing protocol and a routed protocol?

Ans Routed protocol: the protocol that carries actual user data (like IPv4, IPv6). It is what gets routed. Routing protocol: the protocol that helps routers learn routes so they know WHERE to send routed protocol packets (like OSPF, EIGRP, RIP). Analogy: routed protocol = the cargo (packages). Routing protocol = the GPS system telling drivers (routers) which roads to take.

Q76 What is route summarisation?

Ans Route summarisation (or route aggregation) combines multiple specific routes into one summary route, reducing routing table size. Example: 192.168.0.0/24, 192.168.1.0/24, 192.168.2.0/24, 192.168.3.0/24 can be summarised as 192.168.0.0/22. This reduces the number of routes advertised, saving memory and CPU on routers. Smaller routing tables = faster route lookup = better performance.

Q77 What is floating static route?

Ans A floating static route is a backup static route with a higher AD than the primary route. It sits in the background and only activates if the primary route disappears. Example: Primary route via OSPF (AD 110). Backup static route: `ip route 192.168.2.0 255.255.255.0 10.0.0.2 150` (AD 150). If OSPF goes down, the backup static route with AD 150 becomes active. This is a simple way to create route redundancy.

Q78 What is load balancing in routing?

Ans Load balancing distributes traffic across multiple paths to the same destination. Equal-cost load balancing: if two paths have the same metric, the router sends traffic across both (round-robin). OSPF and EIGRP support this by default. EIGRP also supports unequal-cost load balancing using the 'variance' command — traffic is distributed proportionally based on path cost. This maximises bandwidth utilisation.

Q79 What is a next-hop IP address?

Ans The next-hop IP is the IP address of the NEXT router in the path to a destination. When you see a route like: O 192.168.2.0/24 [110/2] via 10.0.0.2, the next-hop is 10.0.0.2. The router forwards the packet to 10.0.0.2, which then forwards it closer to the destination. The next-hop must be directly reachable from the current router (on a connected network).

Q80 What is OSPF neighbour adjacency?

Ans Before OSPF routers can exchange routes, they must become 'neighbours'. Requirements: Both routers must be in the same OSPF area, same subnet, same hello/dead timer intervals, same authentication (if configured), and matching MTU. Check neighbours: show ip ospf neighbor. States: Down □ Init □ 2-Way □ Exstart □ Exchange □ Loading □ Full. 'Full' state means fully synchronised and working correctly.

Q81 What is Cisco's show ip route output?

Ans Reading the routing table: 'O 192.168.2.0/24 [110/2] via 10.0.0.2, 00:01:20, GigabitEthernet0/0' means: O = learned via OSPF, 192.168.2.0/24 = destination network, [110/2] = [AD/metric], via 10.0.0.2 = next-hop IP, 00:01:20 = route age, GigabitEthernet0/0 = exit interface. The router picks the route with lowest AD, then lowest metric.

Q82 What is a stub network?

Ans A stub network is a network connected to only ONE router — there is no other way in or out. Example: a branch office connected to headquarters via a single WAN link. The branch router only needs a default route (0.0.0.0/0) to forward all traffic to HQ. The HQ router knows the specific route to the branch. Understanding stubs helps simplify routing design and configuration.

Q83 What is Cisco Express Forwarding (CEF)?

Ans CEF is Cisco's high-performance packet forwarding mechanism. Instead of the router CPU making a routing decision for every packet (process switching — slow), CEF builds two tables in hardware: FIB (Forwarding Information Base — copy of routing table) and Adjacency Table (Layer 2 next-hop info). Packets are forwarded using hardware — much faster, lower CPU. CEF is enabled by default on modern Cisco platforms.

Q84 What is a routing loop and how is it prevented?

Ans A routing loop occurs when packets bounce between routers indefinitely because each router thinks another is the correct next-hop. Prevention mechanisms: TTL (Time To Live) — each router decrements TTL by 1; packet discarded when TTL=0. Split horizon — do not advertise a route back out the interface it was learned from. Route poisoning — advertise failed route with metric of infinity. Hold-down timers — ignore updates for a period after a route goes down.

Q85 What is OSPF area 0?

Ans In OSPF, all routers are organised into areas. Area 0 is the backbone area — every other area MUST connect to Area 0. Why? This ensures all routes flow through a central point, preventing routing loops between areas. If you have a small network, put all routers in Area 0 (single area OSPF). Multi-area OSPF is used in large networks to reduce routing table size and control LSA flooding.

Q86 What is the difference between interior and exterior gateway protocols?

Ans Interior Gateway Protocol (IGP): routes traffic within a single organisation (Autonomous System). Examples: OSPF, EIGRP, RIP. Exterior Gateway Protocol (EGP): routes traffic between different organisations. Example: BGP (the only EGP used today). Your company uses OSPF internally. Your internet router uses BGP to exchange routes with your ISP. CCNA focuses on IGPs, particularly OSPF and EIGRP.

Q87 What is the passive interface command in OSPF/EIGRP?

Ans The passive-interface command stops a routing protocol from sending routing updates out a specific interface while still advertising that network. Why? An interface connected to end users (PCs, printers) does not need to receive routing updates — it wastes bandwidth and is a security risk. Command: `router ospf 1`
`passive-interface GigabitEthernet 0/1`. The network is still advertised; just no hello packets are sent on that interface.

Q88 How do you verify OSPF is working correctly?

Ans Key verification commands: show ip ospf neighbor (check neighbour adjacency — should show 'Full' state), show ip ospf (process info, Router ID, area), show ip route ospf (routes learned via OSPF), show ip ospf interface (OSPF settings per interface — hello/dead timers, DR/BDR). Common problems: neighbours stuck in INIT (mismatched subnet), DOWN (no connectivity), or EXSTART (MTU mismatch).

Q89 What is EIGRP successor and feasible successor?

Ans In EIGRP, the best route to a destination is the Successor route — stored in routing table. EIGRP also keeps a backup route called the Feasible Successor. If the primary route fails, EIGRP immediately switches to the feasible successor without recalculating — this is why EIGRP converges so fast. The feasible successor must pass the Feasibility Condition: its reported distance must be less than the current successor's feasible distance.

Q90 What is redistribute in routing?

Ans Redistribution allows routes from one routing protocol to be imported into another. Example: your company uses OSPF internally but acquires another company running EIGRP. Redistribution lets OSPF routers learn EIGRP routes and vice versa. Command (on border router): redistribute eigrp 1 subnets (into OSPF). Redistribution can be complex and may cause routing loops if not carefully designed. Metrics must be set correctly when redistributing.

SECTION 04

Switching & STP

Q91 –
Q120

VLANs · Spanning Tree · EtherChannel · Port Security · Layer 3 Switching

Q91 How does a switch learn MAC addresses?

Ans A switch learns MAC addresses by watching incoming frames. When a frame arrives, the switch records the source MAC address and the port it came in on — this is stored in the MAC Address Table (CAM table). Next time a frame arrives for that MAC, the switch sends it only to the correct port instead of flooding to all ports. This is called unicast forwarding. View the table with: show mac address-table.

TIP *Understanding how switches learn MAC addresses is fundamental to understanding switching behaviour.*

Q92 What is MAC flooding attack?

Ans A MAC flooding attack sends thousands of frames with fake MAC addresses to overflow the switch's MAC address table. When the table is full, the switch cannot learn new addresses and starts flooding ALL traffic to ALL ports — like a hub. Attacker can then capture traffic using a packet sniffer. Prevention: Port Security (limits number of MAC addresses per port), 802.1X authentication.

Q93 What is Spanning Tree Protocol (STP)?

Ans STP prevents Layer 2 loops in networks with redundant links. Why are loops bad? A broadcast frame would loop forever between switches — in seconds the network is completely overwhelmed (broadcast storm). STP works by blocking redundant ports. If the active path fails, STP unblocks a blocked port to restore connectivity. Converging after a failure takes 30-50 seconds with classic STP.

TIP *Layer 2 loops = instant network crash. STP prevents this. Understand it deeply for CCNA!*

Q94 What is the STP Root Bridge and how is it elected?

Ans The Root Bridge is the switch that becomes the centre of the STP topology — all other switches calculate their best path toward the Root Bridge. Election: lowest Bridge ID wins. Bridge ID = Priority (default 32768) + MAC address. If all priorities are equal (likely in a lab), the switch with the lowest MAC address wins. Best practice: manually set the important core switch as Root Bridge using 'spanning-tree vlan 1 priority 4096'.

Q95 What are STP port roles?

Ans STP assigns roles to every port: Root Port (RP): port on each non-root switch that is closest (best path) to the Root Bridge — one per switch. Designated Port (DP): port that forwards traffic away from the Root Bridge on each network segment — one per segment. Blocked Port (BP): any port that is not RP or DP — blocked to prevent loops. Root Bridge ports are all Designated Ports.

Q96 What are STP port states?

Ans STP ports go through states: Blocking (receives BPDUs but does not forward data — 20 seconds), Listening (sends/receives BPDUs, not forwarding data — 15 seconds), Learning (building MAC table, not forwarding data — 15 seconds), Forwarding (fully operational — forwards data). Disabled (administratively shut down). Total time from blocking to forwarding = 50 seconds — this is STP convergence time.

Q97 What is RSTP (Rapid Spanning Tree Protocol)?

Ans RSTP (IEEE 802.1w) is the modern replacement for STP. It converges much faster — typically 1-2 seconds instead of 30-50 seconds. RSTP adds new port roles (Alternate and Backup) and port states (Discarding, Learning, Forwarding). It uses a negotiation mechanism between switches (proposal/agreement) for rapid transition. Modern Cisco switches run RSTP by default. Configure: spanning-tree mode rapid-pvst.

Q98 What is PortFast and BPDU Guard?

Ans PortFast: skips STP listening and learning states on access ports connected to end devices (PCs, printers). Port goes directly to forwarding — eliminates the 30-50 second wait. Configure: spanning-tree portfast on the interface. BPDU Guard: if a PortFast port receives a BPDU (from another switch), it immediately shuts the port down (err-disabled). Prevents someone plugging a rogue switch into an access port and disrupting STP.

Q99 What is EtherChannel?

Ans EtherChannel (also called Link Aggregation or port bonding) bundles multiple physical links between two switches into one logical link. Benefits: increased bandwidth (4 x 1Gbps = 4Gbps logical), redundancy (if one physical link fails, others keep working), STP sees it as one link (no blocking). Protocols: LACP (802.3ad, open standard — preferred) and PAgP (Cisco proprietary). Configure: channel-group 1 mode active (LACP).

Q100 What is a Layer 3 switch?

Ans A Layer 3 switch can perform both switching (Layer 2) and routing (Layer 3). It routes between VLANs using SVIs (Switch Virtual Interfaces) – virtual interfaces with IP addresses. Much faster than Router-on-a-Stick (ROAS) because routing is done in hardware ASICs. Configure an SVI: interface vlan 10 □ ip address 192.168.10.1 255.255.255.0 □ no shutdown. Enable routing: ip routing. Best choice for inter-VLAN routing in enterprise networks.

Q101 What is a SVI (Switch Virtual Interface)?

Ans SVI is a virtual interface on a Layer 3 switch representing a VLAN. It acts as the default gateway for devices in that VLAN. Create it: interface vlan 10 □ ip address 192.168.10.1 255.255.255.0 □ no shutdown. The SVI will only come up if: the VLAN exists in the VLAN database, and at least one active port is assigned to that VLAN. SVIs are also used for switch management access.

Q102 What is port security?

Ans Port security limits which devices can connect to a switch port based on MAC address. Configuration: switchport mode access □ switchport port-security □ switchport port-security maximum 1 □ switchport port-security violation shutdown. Modes when violation occurs: Shutdown (disables port – most secure, must manually re-enable), Restrict (drops unauthorised frames, sends SNMP alert), Protect (drops unauthorised frames silently).

Q103 What is a trunk allowed VLAN list?

Ans By default, a trunk port carries ALL VLANs (1-4094). Best practice: restrict trunks to only the VLANs that actually need to traverse that link. Command: switchport trunk allowed vlan 10,20,30. This reduces unnecessary traffic and improves security (VLAN 40 traffic cannot cross a trunk that does not include VLAN 40). Verify: show interfaces trunk.

Q104 What is the 'show vlan brief' command?

Ans This command shows all VLANs configured on the switch and which ports belong to each VLAN. Output: VLAN number, name, status (active/suspended), and ports assigned. Useful for: verifying VLAN configuration, checking that ports are in the correct VLAN, troubleshooting VLAN issues. Note: trunk ports do NOT appear in 'show vlan brief' – only access ports are listed per VLAN.

Q105 What is VLAN database vs global config VLAN?

Ans There are two ways to create VLANs on Cisco switches. Older method: vlan database mode (type 'vlan database' from privileged mode). Modern method (recommended): create VLAN in global config mode (vlan 10 name Sales). Both create VLANs. On Layer 2 switches, VLAN info is stored in flash:vlan.dat file — this is why you must delete vlan.dat when erasing a switch's config.

Q106 What is PVST+ (Per-VLAN Spanning Tree Plus)?

Ans Cisco's PVST+ runs a separate STP instance for each VLAN. Benefits: you can have different root bridges for different VLANs — enabling load balancing across redundant links. Example: VLAN 10 root bridge = Switch A, VLAN 20 root bridge = Switch B. Both uplinks are used simultaneously (just for different VLANs) instead of one being blocked. Configure: spanning-tree vlan 10 priority 4096.

Q107 What is the MAC address table and how does it age out?

Ans The MAC address table (CAM table) stores MAC-to-port mappings. Default aging time: 300 seconds (5 minutes). If a device goes quiet for 5 minutes, its entry is removed. Next time it sends traffic, the switch must flood to relearn its location. You can view the table: show mac address-table. You can adjust aging: mac address-table aging-time 600. Static entries never age out: mac address-table static [mac] vlan [id] interface [port].

Q108 What is BPDUFilter?

Ans BPDUFilter stops a port from sending or receiving BPDUs entirely. When applied globally with PortFast, it only filters BPDUs on PortFast-enabled ports that are not receiving BPDUs. When applied per-interface, it completely disables STP on that interface — dangerous! If a loop exists, it will not be detected. Use with extreme caution. BPDUGuard is safer for access ports. Only use BPDUFilter when you truly need to disable STP on a specific interface.

Q109 What is storm control on a switch?

Ans Storm control limits the amount of broadcast, multicast, or unknown unicast traffic on a port. If traffic exceeds the threshold, the switch drops excess packets or shuts down the port. This protects against broadcast storms (caused by loops, STP failure, or network attacks). Configure: storm-control broadcast level 20 (allow max 20% broadcast traffic). Essential in production networks as extra protection alongside STP.

Q110 How do you troubleshoot VLAN issues?

Ans VLAN troubleshooting checklist: 1) Is VLAN created on BOTH switches? (show vlan brief). 2) Is the access port in the correct VLAN? (show interfaces [port] switchport). 3) Does the trunk carry the VLAN? (show interfaces trunk). 4) Does the Layer 3 switch have an SVI for the VLAN? (show ip interface brief). 5) Is the SVI up? (it needs an active port in that VLAN). 6) Is the default gateway set correctly on end devices?

Q111 What is native VLAN mismatch?

Ans A native VLAN mismatch occurs when the two ends of a trunk link have different native VLANs configured. Example: Switch A native VLAN = 1, Switch B native VLAN = 99. Symptoms: some traffic crosses the wrong VLAN, Cisco switches generate CDP warnings ('Native VLAN mismatch discovered'). This is both a connectivity problem and a security vulnerability (VLAN hopping). Fix: configure the same native VLAN on both ends.

Q112 What is DTP (Dynamic Trunking Protocol)?

Ans DTP is Cisco's proprietary protocol that automatically negotiates trunk formation between connected switches. Modes: Dynamic Auto (forms trunk if other end is Dynamic Desirable or Trunk), Dynamic Desirable (actively tries to form trunk), Trunk (always trunk), Access (never trunk). Security best practice: disable DTP on access ports (switchport nonegotiate) to prevent rogue switches from negotiating a trunk automatically.

Q113 What is VTP (VLAN Trunking Protocol)?

Ans VTP allows VLAN configuration to be synchronised across multiple switches automatically. Modes: Server (can create/delete VLANs, propagates changes), Client (receives VLAN updates, cannot create VLANs), Transparent (does not participate in VTP but forwards VTP messages). Warning: connecting an old switch with higher VTP revision number can ERASE all VLANs on your network! Best practice for small networks: use VTP transparent or disable VTP.

Q114 What is a root guard in STP?

Ans Root Guard prevents a port from becoming a root port. If a superior BPDU is received on a root guard-enabled port, the port is placed in a 'root-inconsistent' state (blocked). This prevents a rogue or misconfigured switch from taking over as Root Bridge and disrupting the STP topology. Enable on ports connected toward access/distribution switches: spanning-tree guard root. Complements BPDU Guard which is for end-device ports.

Q115 What is the difference between cut-through and store-and-forward switching?

Ans Store-and-forward: switch receives the entire frame, checks for errors (CRC), then forwards. Catches corrupted frames but adds latency. Most common mode. Cut-through: switch reads only the destination MAC (first 6 bytes) and immediately starts forwarding before receiving the full frame. Faster but forwards corrupted frames. Fragment-free: reads first 64 bytes (checks for collisions) then forwards — compromise between both.

Q116 What is a unidirectional link?

Ans A unidirectional link occurs when only one direction of a fibre pair is working — the switch thinks the link is up but traffic only flows one way. This can create STP issues (switch thinks a link is working and does not block it, creating a loop). Cisco's UDLD (UniDirectional Link Detection) protocol detects and disables unidirectional links. Enable: udld enable (globally) or udld port (per interface).

Q117 What is 802.1X on switch ports?

Ans 802.1X provides port-based network access control. Before a device can use the network, it must authenticate. Components: Supplicant (client device), Authenticator (switch), Authentication Server (RADIUS). If authentication fails, the port is placed in a guest VLAN or denied access entirely. This prevents unauthorized devices from connecting. Configure: aaa new-model, dot1x system-auth-control, then per-interface: dot1x port-control auto.

Q118 What is the purpose of the 'spanning-tree portfast default' command?

Ans This global command enables PortFast on ALL access mode interfaces automatically. You do not need to configure it port-by-port. It only applies to ports in access mode — trunk ports are not affected. Used in environments where all access ports connect to end devices (PCs, printers, phones). Combined with 'spanning-tree portfast bpduguard default' which also enables BPDU Guard globally on all PortFast ports.

Q119 What is a Layer 2 loop and how does it cause a broadcast storm?

Ans A Layer 2 loop happens when there are multiple paths between switches and STP is not running. What happens: Device A sends a broadcast (e.g. ARP request). Switch 1 floods it out all ports including both uplinks to Switch 2. Switch 2 receives on both ports and floods out all ports — sending it back to Switch 1. Repeat endlessly. In milliseconds, millions of frames circulate — completely saturating every link and crashing the network.

Q120 How do you verify EtherChannel is working?

Ans Verification commands: show etherchannel summary (shows bundle status — P = in port-channel, I = stand-alone problem). show etherchannel detail (detailed stats). show interfaces port-channel 1 (logical interface status). Common problems: mismatched duplex/speed, different VLANs on ports, different STP settings, one side using LACP and other using PAgP. All ports in an EtherChannel must have identical configuration.

SECTION 05	IP Services	Q121 – Q150
NAT · DHCP · NTP · SNMP · Syslog · QoS · HSRP · FHRP		
Q121	What is DHCP and how do you configure it on a Cisco router?	
Ans	DHCP server config on Cisco: <code>ip dhcp pool LAN_POOL</code> □ <code>network 192.168.1.0 255.255.255.0</code> □ <code>default-router 192.168.1.1</code> □ <code>dns-server 8.8.8.8</code> □ <code>lease 7 (days)</code>. Exclude addresses you want static: <code>ip dhcp excluded-address 192.168.1.1 192.168.1.10</code>. Verify: <code>show ip dhcp binding</code> (see all leased IPs), <code>show ip dhcp pool</code> (pool stats). Router acts as DHCP server for all devices on the connected LAN.	
TIP	CCNA often tests DHCP configuration and troubleshooting steps!	
Q122	What is a DHCP relay agent?	
Ans	By default, DHCP uses broadcasts which do not cross routers. If your DHCP server is on a different subnet than clients, configure a DHCP relay agent. On the router interface facing the clients, type: <code>ip helper-address [DHCP-server-IP]</code>. The router will convert the client's DHCP broadcast into a unicast message directed at the DHCP server. This allows one central DHCP server to serve multiple subnets across a routed network.	
Q123	What are the three types of NAT?	
Ans	Static NAT: one private IP permanently maps to one public IP. Used for servers that need to be reachable from the internet. Dynamic NAT: maps a private IP to any available public IP from a pool. Overloaded NAT / PAT (Port Address Translation): many private IPs share ONE public IP, distinguished by unique port numbers. PAT is what your home router uses — 20 devices sharing one public IP. PAT is the most common type in real networks.	
Q124	How do you configure PAT (NAT Overload) on a Cisco router?	
Ans	Steps: Create ACL identifying inside addresses: <code>access-list 1 permit 192.168.1.0 0.0.0.255</code>. Configure NAT: <code>ip nat inside source list 1 interface GigabitEthernet0/1 overload</code>. Mark interfaces: <code>int g0/0</code> □ <code>ip nat inside</code>. <code>int g0/1</code> □ <code>ip nat outside</code>. Verify: <code>show ip nat translations</code> (see active NAT entries), <code>show ip nat statistics</code>. The 'overload' keyword enables PAT.	

Q125 What is NTP (Network Time Protocol)?

Ans NTP synchronises clocks across all network devices. Why it matters: log files from different devices must have matching timestamps for troubleshooting and security investigation. Kerberos authentication (used by Active Directory) fails if clocks differ by more than 5 minutes. Configure Cisco to sync with internet NTP server: `ntp server 216.239.35.0`. Verify: `show ntp status`, `show clock`. NTP uses UDP port 123. Stratum 1 = connected to atomic clock, Stratum 2 = synced from Stratum 1.

Q126 What is SNMP and what is it used for?

Ans SNMP (Simple Network Management Protocol) is used to monitor and manage network devices remotely. An NMS (Network Management Station) polls devices for stats like CPU, memory, bandwidth. Devices can also send Traps to the NMS when something goes wrong (port down, high CPU). SNMP versions: v1 and v2c use community strings (like passwords — 'public' for read, 'private' for write). SNMPv3 is secure — uses authentication and encryption. Use SNMPv3 in production.

Q127 What is Syslog?

Ans Syslog sends log messages from network devices to a central syslog server. Why? Logs are vital for troubleshooting and security. Storing logs on the device itself is limited (buffer fills up). A syslog server keeps all logs from all devices in one place. Configure: `logging host 192.168.1.100`. Logging levels 0-7: 0=Emergencies (worst), 1=Alerts, 2=Critical, 3=Errors, 4=Warnings, 5=Notifications, 6=Informational, 7=Debugging (most verbose). Syslog uses UDP port 514.

Q128 What is HSRP (Hot Standby Router Protocol)?

Ans HSRP provides gateway redundancy. If you have two routers connecting your LAN to the internet, PCs need to point to ONE default gateway IP. HSRP creates a virtual IP address shared between two routers. One is Active (handles traffic), one is Standby (monitoring). If Active fails, Standby takes over the virtual IP automatically — PCs notice nothing. Configure: `standby 1 ip 192.168.1.254` □ `standby 1 priority 110` (higher wins Active role).

TIP

HSRP = high availability for gateways. Know Active/Standby election and virtual IP concept for CCNA!

Q129 What is VRRP?

Ans VRRP (Virtual Router Redundancy Protocol) is the open-standard equivalent of HSRP — works on all vendors, not just Cisco. One router is the Master (equivalent to HSRP Active), others are Backup. Configured the same way conceptually. `vrrp 1 ip 192.168.1.254` `vrrp 1 priority 200`. Both HSRP and VRRP achieve the same goal: transparent gateway failover without end users needing to change their default gateway setting.

Q130 What is GLBP (Gateway Load Balancing Protocol)?

Ans GLBP is Cisco's improvement over HSRP. While HSRP has one Active router handling ALL traffic (standby is idle), GLBP distributes traffic across ALL routers simultaneously. Each router is an AVF (Active Virtual Forwarder) with its own virtual MAC address. An AVG (Active Virtual Gateway) assigns different virtual MACs to different clients. Result: all routers carry traffic = full bandwidth utilisation. Cisco proprietary.

Q131 What is QoS (Quality of Service)?

Ans QoS prioritises certain types of traffic to ensure critical applications perform well even when the network is congested. Without QoS, all traffic is treated equally — a large file download could ruin a VoIP call. With QoS: VoIP packets get priority (low latency, low jitter). Video conferencing gets guaranteed bandwidth. Bulk downloads are deprioritised. Mark traffic with DSCP values. Queue management ensures high-priority traffic goes first.

Q132 What is traffic marking in QoS?

Ans Traffic must be identified and marked before it can be prioritised. DSCP (Differentiated Services Code Point): 6-bit field in IP header. Common values: EF (Expedited Forwarding, 46) = VoIP, AF41 (34) = video, CS0 = best effort (default). CoS (Class of Service): 3-bit field in 802.1Q VLAN tag (Layer 2). Classification: use ACLs or NBAR (Network Based Application Recognition) to identify traffic and mark it appropriately.

Q133 What is the difference between CBWFQ and LLQ?

Ans CBWFQ (Class-Based Weighted Fair Queuing): assigns guaranteed bandwidth to different traffic classes. Good for data traffic. LLQ (Low Latency Queuing): adds a strict priority queue to CBWFQ. The strict priority queue is served first before all other queues. Best for VoIP — VoIP packets get out immediately, meeting the low latency (< 150ms) and low jitter (<30ms) requirements for good voice quality.

Q134 What is a DHCP conflict?

Ans A DHCP conflict occurs when two devices have the same IP address — one from DHCP, one manually configured. The DHCP server detects this by sending a ping before assigning an IP. If the ping gets a reply, the address is already in use and moved to a conflict list. Command: `show ip dhcp conflict`. Fix: exclude the manually-assigned IPs from the DHCP pool using `'ip dhcp excluded-address'`.

Q135 What is CDP in IP services context?

Ans CDP (Cisco Discovery Protocol) helps gather neighbour information — IP addresses, platform, port connections. Useful for building or verifying network maps. In IP services context, CDP advertisements contain the device's IP address which can be used for management access. Security: disable CDP on user-facing ports and internet-facing ports to avoid leaking device information. Global disable: `no cdp run`.

Q136 What is SNMP community string?

Ans A community string is like a password for SNMP v1 and v2c. Read-only community (default 'public'): NMS can read device data. Read-write community (default 'private'): NMS can also change device configuration. Never use 'public' or 'private' defaults — change them! Configure: `snmp-server community MyReadOnly ro`, `snmp-server community MyReadWrite rw`. Better: use SNMPv3 with proper usernames and encrypted passwords.

Q137 What is IP SLA on Cisco routers?

Ans IP SLA (Service Level Agreement) allows a Cisco router to generate test traffic and measure network performance metrics: latency, jitter, packet loss, availability. Uses: verify WAN link quality, automatically switch to backup route when primary degrades (with floating static route and tracking). Configure: `ip sla 1` `icmp-echo 8.8.8.8` `frequency 10` `ip sla schedule 1 start-time now life forever`. Verify: `show ip sla statistics`.

Q138 What is HSRP preempt?

Ans By default in HSRP, if the Active router fails and the Standby takes over, the original Active router does NOT take back the Active role when it recovers — the Standby remains Active. The `'standby 1 preempt'` command changes this: the router with higher priority WILL reclaim the Active role automatically when it comes back online. Use preempt when you want a specific router to always be Active.

Q139 What is the difference between unicast, multicast and broadcast for DHCP?

Ans DHCP Discover and Request: sent as broadcasts because the client has no IP yet. DHCP Offer and Acknowledge: server sends unicast to client's MAC (or broadcast if client sets broadcast flag). DHCP renewals (halfway through lease): sent as unicast directly to DHCP server IP. DHCP release: sent as unicast. This is why DHCP relay is needed — broadcasts cannot cross routers but relay converts them to unicast.

Q140 What is NetFlow?

Ans NetFlow collects information about IP traffic flowing through a router — who talked to whom, how much data, which application (port). This data is exported to a NetFlow collector for analysis. Uses: bandwidth monitoring (see top talkers), security analysis (detect unusual traffic patterns), billing (how much bandwidth each user consumed), capacity planning. Enable: ip flow ingress on interface. Export: ip flow-export destination [collector IP] [port].

Q141 What is FHRP (First Hop Redundancy Protocol)?

Ans FHRP is the category of protocols providing gateway redundancy. Types: HSRP (Cisco, hot standby), VRRP (open standard, virtual router), GLBP (Cisco, load balancing). All provide a virtual IP address that end devices use as default gateway. When the active router fails, another router takes over the virtual IP seamlessly. Without FHRP, if the gateway router fails, all devices lose internet access until manually reconfigured.

Q142 What is object tracking with HSRP?

Ans HSRP can track an interface or IP SLA object and automatically change priority based on the tracked object's state. Example: track the uplink to the ISP. If the uplink goes down, reduce HSRP priority by 20 — causing the other router (which still has a working uplink) to become Active. Configure: track 1 interface G0/1 line-protocol □ standby 1 track 1 decrement 20. This ensures the active HSRP router always has a working path to the internet.

Q143 What is a DNS server hierarchy?

Ans DNS uses a hierarchical structure: Root DNS servers (13 sets worldwide, managed by ICANN — know the location of all TLD servers). TLD servers (for .com, .org, .in — know the authoritative nameservers for each domain). Authoritative DNS servers (know the actual IP for a specific domain). Recursive resolver (your ISP or Google DNS 8.8.8.8 — does the work of asking all the above on your behalf). This hierarchy makes DNS scalable and resilient.

Q144 What is the purpose of logging synchronous?

Ans When Cisco IOS sends log messages to the console, they can appear in the middle of what you are typing — very annoying! The 'logging synchronous' command tells IOS to redisplay your partially-typed command on a new line after a log message appears. Configure: `line con 0` `logging synchronous`. Also useful on VTY lines. Makes working on the CLI much more comfortable, especially in busy networks with lots of messages.

Q145 What is the exec-timeout command?

Ans `exec-timeout [minutes] [seconds]` sets the inactivity timeout for console and VTY sessions. After this time of no activity, the session closes. Configure: `line con 0` `exec-timeout 5 0` (5 minutes). Security best practice: set a timeout so unattended sessions auto-close. 'exec-timeout 0 0' disables timeout (no auto-disconnect — useful in lab but dangerous in production). Default is 10 minutes on most platforms.

Q146 What is CDP holdtime?

Ans CDP holdtime is how long a device keeps CDP information from a neighbour before discarding it (if no new CDP advertisement is received). Default holdtime = 180 seconds. Default CDP send timer = 60 seconds. So a neighbour entry is kept for 3 x the send interval by default. Change: `cdp holdtime 120` or `cdp timer 30`. CDP runs on Layer 2 and is sent to multicast address 01:00:0C:CC:CC:CC.

Q147 What is IPv6 Stateless Address Autoconfiguration (SLAAC)?

Ans SLAAC allows IPv6 devices to automatically configure their own address without a DHCP server. Process: device generates a link-local address (FE80:: + modified MAC address). Device listens for Router Advertisements (RA) from the router. RA contains the network prefix (e.g. 2001:db8::/64). Device combines prefix + its interface ID to form a global address. No DHCP server needed! IPv6 also has DHCPv6 for more control.

Q148 What is ICMP and what is it used for?

Ans ICMP (Internet Control Message Protocol) carries error messages and operational information about IP. Common uses: Ping (Echo Request/Reply — test connectivity), Traceroute (Time Exceeded messages), Destination Unreachable (no route to host, port closed), Redirect (better path available). ICMP is not used to carry user data. Firewalls sometimes block ICMP — so a failed ping does not always mean a host is down.

Q149 What is a GRE tunnel?

Ans GRE (Generic Routing Encapsulation) creates a virtual point-to-point tunnel between two routers over an IP network. It wraps (encapsulates) original packets inside new IP packets. Uses: connect two private networks over the internet (like a basic VPN), carry non-IP protocols, route multicast over networks that do not support it. GRE itself has no encryption — combine with IPsec for a secure VPN. Configure: `interface tunnel 0` □ `tunnel source` □ `tunnel destination` □ `ip address`.

Q150 What is DHCP starvation?

Ans DHCP starvation is an attack where a malicious device sends thousands of DHCP Discover messages with fake MAC addresses, consuming all available IP addresses in the DHCP pool. Legitimate devices then cannot get an IP address. Prevention: DHCP Snooping on switches — limits the number of DHCP requests per port and validates DHCP messages. Enable: `ip dhcp snooping (global)` □ `ip dhcp snooping vlan 10` □ `no ip dhcp snooping information option` (if using basic config).

SECTION 06

Security Fundamentals

Q151 –
Q180

ACLs · AAA · Threats · VPN · Wireless Security · Device Hardening · Firewalls

Q151 What is network security and why does it matter?

Ans Network security protects data, devices, and systems from unauthorised access, attacks, and damage. Without security: attackers can steal passwords and data, ransomware can encrypt your files, someone can disrupt your entire network. Good security uses multiple layers: physical security (locks), network security (firewalls, ACLs, VPN), device security (strong passwords, SSH), and user awareness (training against phishing).

TIP Security is ~15% of CCNA exam. Understand concepts even if you do not configure every technology.

Q152 What is an Access Control List (ACL)?

Ans An ACL is a list of rules on a router that permits or denies traffic based on IP addresses and ports. Think of it as a bouncer at a club — checks each packet and decides whether to let it pass. Standard ACL: filters by source IP only. Extended ACL: filters by source IP, destination IP, protocol, and port number. Rules are checked top-down — first match wins. There is always an implicit 'deny all' at the end.

Q153 What is the difference between standard and extended ACLs?

Ans Standard ACL (numbered 1-99, 1300-1999): only filters on source IP address. Simple but limited. Place close to destination (because it blocks based only on source — placing near source would block too broadly). Extended ACL (numbered 100-199, 2000-2699): filters on source IP, destination IP, protocol (TCP/UDP/ICMP), and port numbers. Much more specific. Place close to source to stop traffic early and save bandwidth.

Q154 How do you configure a standard named ACL?

Ans Create ACL: `ip access-list standard BLOCK_SALES` □ `deny 192.168.10.0 0.0.0.255` □ `permit any`. Apply to interface: `interface g0/0` □ `ip access-group BLOCK_SALES in` (or out). 'in' = filter traffic coming INTO the interface from that direction. 'out' = filter traffic going OUT. Named ACLs are easier to edit than numbered ACLs. Verify: `show access-lists`.

Q155 What is wildcard mask in an ACL?

Ans Wildcard mask is the inverse of a subnet mask. 0 means must match, 1 means do not care. Examples: 0.0.0.255 matches any host in a /24 network (last octet can be anything). 0.0.0.0 matches exactly ONE host. 255.255.255.255 matches ANY address ('any' keyword). Subnet mask 255.255.255.0 □ wildcard 0.0.0.255. Simple calculation: subtract subnet mask from 255.255.255.255.

Q156 What is AAA in network security?

Ans AAA stands for Authentication (who are you?), Authorisation (what are you allowed to do?), and Accounting (what did you do?). Centralised AAA uses a RADIUS or TACACS+ server to manage all three for network device access. Example: IT engineer logs in to a switch □ RADIUS server authenticates their username/password, authorises which commands they can use, and logs every command typed. Much more scalable than local usernames on every device.

Q157 What is the difference between RADIUS and TACACS+?

Ans RADIUS: open standard (RFC), uses UDP 1812/1813, combines authentication and authorisation (less granular), encrypts only the password. Best for: network access (Wi-Fi, VPN users). TACACS+: Cisco proprietary, uses TCP 49, separates authentication, authorisation, and accounting (more granular — can restrict specific commands), encrypts the entire packet. Best for: managing network devices (routers, switches). Both use a centralised server for AAA.

Q158 What is a firewall and how does it differ from an ACL?

Ans A firewall is a dedicated security device (hardware or software) that controls traffic between networks. ACL: simple rule-based filtering on a router, stateless (does not track connections). Firewall: stateful — tracks active TCP connections. Knows if a packet is part of an established session or a new suspicious connection. Can block return traffic from unsolicited connections automatically. Modern firewalls also do application-layer inspection, URL filtering, and IPS.

Q159 What are common network attack types?

Ans DoS (Denial of Service): flood a device with traffic to make it unavailable. DDoS: same but from thousands of infected computers. Phishing: fake emails tricking users into revealing passwords. Man-in-the-Middle (MITM): attacker intercepts communication between two devices. ARP Spoofing: attacker sends fake ARP replies to redirect traffic through their device. Password attacks: brute force (try every combination) or dictionary attacks. Ransomware: encrypts your files and demands payment.

Q160 What is port security and why is it a security feature?

Ans Port security prevents unauthorised devices from connecting to switch ports. An attacker plugging into a network port should not automatically get network access. Port security limits which MAC addresses can use a port. If an unauthorised device connects, the port shuts down or drops its traffic. Sticky learning automatically learns the first device's MAC and locks the port to it — great for offices where desks do not change.

Q161 What is DHCP snooping?

Ans DHCP snooping protects against rogue DHCP servers. An attacker can set up a fake DHCP server and give clients wrong IPs, wrong gateways (pointing to the attacker), or wrong DNS — enabling Man-in-the-Middle attacks. DHCP snooping: marks switch ports as Trusted (connected to real DHCP server) or Untrusted (connected to client devices). DHCP Offer and Acknowledge messages received on Untrusted ports are dropped.

Q162 What is Dynamic ARP Inspection (DAI)?

Ans ARP spoofing/poisoning sends fake ARP replies to redirect traffic. DAI validates ARP packets against the DHCP snooping binding table (which records correct IP-MAC-port mappings). If an ARP packet's IP-MAC mapping does not match the binding table, it is dropped. DAI requires DHCP snooping to be enabled first (to build the binding table). Enable: `ip arp inspection vlan 10`.

Q163 What is 802.1X network access control?

Ans 802.1X controls network access at the switch port level. Before traffic is allowed, the connecting device must authenticate. Three roles: Supplicant (device wanting access), Authenticator (the switch), Authentication Server (RADIUS). If authentication fails: port goes to Guest VLAN (limited access) or stays blocked. Used in enterprise environments to ensure only company-owned, domain-joined devices access the network.

Q164 What is a DMZ (Demilitarized Zone)?

Ans A DMZ is a separate network zone sitting between your internal network and the internet. Public-facing servers (web server, email server, DNS) are placed in the DMZ. Why? If an attacker compromises your web server, they are in the DMZ — not directly in your internal network where employee PCs and databases live. The firewall has strict rules: internet can reach DMZ servers, but DMZ cannot freely access the internal network.

Q165 What is VPN (Virtual Private Network)?

Ans A VPN creates an encrypted tunnel over the internet, allowing secure communication. Two main types: Remote Access VPN: employee working from home connects securely to company network as if in the office. Client software (Cisco AnyConnect) creates the tunnel. Site-to-Site VPN: connects two offices over the internet securely — replaces expensive dedicated WAN links. IPsec provides the encryption. Both types protect data from interception.

Q166 What is IPsec?

Ans IPsec (Internet Protocol Security) is a suite of protocols that encrypts and authenticates IP packets. Components: AH (Authentication Header): ensures data has not been tampered with. ESP (Encapsulating Security Payload): encrypts data + ensures integrity. IKE (Internet Key Exchange): negotiates the encryption keys between two ends. Two modes: Transport (encrypts only payload), Tunnel (encrypts entire packet — used in VPNs). IPsec runs over UDP 500 and 4500.

Q167 What is wireless security WPA2 vs WPA3?

Ans WPA2: uses AES encryption (much better than old WEP/TKIP). Two modes: Personal (pre-shared key — home/small office), Enterprise (802.1X with RADIUS server — corporate). Vulnerability: offline dictionary attacks against the 4-way handshake. WPA3: fixes WPA2 weaknesses. Uses SAE (Simultaneous Authentication of Equals) — protects against offline attacks. Forward secrecy: capturing old traffic is useless even if key is later compromised. WPA3 is the current standard.

Q168 What are the threats to wireless networks?

Ans Evil Twin: attacker creates a fake Wi-Fi hotspot with the same SSID as the real network — victims connect to attacker's AP. Rogue AP: unauthorized access point installed on company network. Deauthentication attack: flood client with fake disconnect packets. Packet sniffing: capture unencrypted wireless traffic. Prevention: use WPA3, 802.1X authentication, wireless IDS to detect rogue APs, VPN on public Wi-Fi.

Q169 What is the principle of least privilege?

Ans The principle of least privilege means giving users and systems only the minimum access they need to do their job — nothing more. Example: a junior IT technician should not have admin rights to all systems. A web server should not have access to the HR database. This limits the damage if an account is compromised. In Cisco: use privilege levels to restrict which commands different users can execute.

Q170 What are Cisco privilege levels?

Ans Cisco IOS has 16 privilege levels (0-15). Level 0: very basic (only logout, enable, disable). Level 1: User EXEC (default — show commands limited). Level 15: Privileged EXEC (all commands). You can create custom levels: privilege exec level 5 show running-config (gives level 5 users access to this command). Assign level to user: username admin privilege 15 secret cisco. Assign level to VTY: privilege level 5.

Q171 What is a Cisco Zone-Based Firewall?

Ans Zone-Based Firewall (ZBF) is a Cisco router firewall feature. You assign interfaces to zones (e.g., inside, outside, DMZ). Traffic policies define what is allowed between zones. By default, traffic between zones is denied — you must explicitly permit it. Traffic within the same zone is allowed by default. More flexible and readable than classic IOS firewall (CBAC). Used when you need firewall features on a router without a dedicated firewall appliance.

Q172 What is crypto key generate rsa?

Ans This command generates RSA public/private key pairs needed for SSH encryption. You must run it before SSH can work. 'crypto key generate rsa modulus 2048' — the modulus is the key size in bits. 2048 bits is recommended for security. Without generating keys, 'ip ssh version 2' has no keys to use. Delete keys: 'crypto key zeroize rsa'. View keys: 'show crypto key mypubkey rsa'.

Q173 What is password hashing and why is it important?

Ans Hashing converts a password into a fixed string that cannot be reversed. In Cisco, 'enable secret' uses MD5 hash — even if someone reads the config, they see 'enable secret 5 \$1\$mERr...' not the actual password. Type 5 is MD5 (older). Type 8 is PBKDF2-SHA256 (stronger, newer). Type 9 is scrypt (strongest, current recommendation). Never use 'enable password' — it stores password in cleartext or Type 7 (easily reversible).

Q174 What is SSH key-based authentication?

Ans Instead of a password, SSH can authenticate using cryptographic key pairs. You generate a key pair on your laptop (private key stays with you, public key copied to the router). When connecting, your private key proves your identity — no password needed. Much more secure than password authentication. Configure on Cisco: ip ssh pubkey-chain □ username admin □ key-string [paste public key]. Disable password auth after setting up key auth.

Q175 What is an IP spoofing attack?

Ans IP spoofing means forging the source IP address in a packet to pretend to be someone else. Why? Bypass IP-based access control, make attacks harder to trace, reflect/amplify attacks (send request with victim's IP as source — all replies hit the victim). Prevention: uRPF (Unicast Reverse Path Forwarding) — router checks if the source IP of an incoming packet matches a route out the same interface. If not, it is likely spoofed and dropped.

Q176 What is Cisco IOS firewall (CBAC)?

Ans CBAC (Context-Based Access Control) is an older Cisco stateful firewall feature. It inspects outbound traffic and dynamically creates temporary ACL entries allowing the corresponding reply traffic back in. Without CBAC, you would need an explicit ACL rule for every possible reply. CBAC understands application protocols (FTP, SMTP, HTTP) and manages dynamic ports correctly. Largely replaced by Zone-Based Firewall in modern networks.

Q177 What is SNMPv3 and why is it preferred?

Ans SNMPv3 adds proper security to SNMP. Three security modes: noAuthNoPriv (no authentication, no encryption — still insecure), authNoPriv (username + password/MD5 authentication but no encryption), authPriv (authentication + encryption — fully secure). Configure: `snmp-server group MyGroup v3 priv` or `snmp-server user admin MyGroup v3 auth sha MyAuthPass priv aes 128 MyEncryptPass`. Always use authPriv in production.

Q178 What is network segmentation and why is it a security control?

Ans Segmenting a network into smaller zones limits how far an attacker can move if they compromise one device. Example: user PCs in one VLAN, servers in another, management devices in a third. A firewall between zones controls what can talk to what. If ransomware infects a PC in the user VLAN, it cannot automatically reach the server VLAN because the firewall blocks it. This is called limiting 'lateral movement'.

Q179 What is two-factor authentication (2FA)?

Ans 2FA requires two different types of verification to prove your identity. Something you know (password) + something you have (phone with authentication app) or something you are (fingerprint). Even if an attacker steals your password, they cannot log in without your phone. CCNA context: Cisco ACS or ISE can enforce 2FA. VPN clients can require password + hardware token. 2FA dramatically reduces account compromise risk.

Q180 What is the CVSS score?

Ans CVSS (Common Vulnerability Scoring System) rates the severity of security vulnerabilities from 0-10. Score ranges: 0.0 = None, 0.1-3.9 = Low, 4.0-6.9 = Medium, 7.0-8.9 = High, 9.0-10.0 = Critical. CCNA security awareness topic. When a new vulnerability is found in Cisco IOS, Cisco publishes an advisory with the CVSS score so network engineers can prioritise which patches to apply first. Higher score = patch urgently.

SECTION 07

Wireless Networking

Q181 –
Q210

Wi-Fi Standards · WLAN Architecture · Channels · Security · Troubleshooting

Q181 What is wireless networking and how does Wi-Fi work?

Ans Wi-Fi uses radio waves to transmit data between devices without cables. Your device (laptop, phone) has a wireless NIC that sends and receives radio signals. A Wireless Access Point (WAP) connects these wireless clients to the wired network. Wi-Fi follows IEEE 802.11 standards. Unlike wired Ethernet (full-duplex), Wi-Fi is half-duplex — devices take turns transmitting and must avoid collisions using CSMA/CA.

TIP CCNA wireless covers WLAN architecture, deployment modes, and basic configuration.

Q182 What are the Wi-Fi 802.11 standards?

Ans 802.11b: 2.4 GHz, max 11 Mbps (very old). 802.11a: 5 GHz, max 54 Mbps (old). 802.11g: 2.4 GHz, max 54 Mbps (common in 2000s). 802.11n (Wi-Fi 4): 2.4 and 5 GHz, up to 600 Mbps, introduced MIMO. 802.11ac (Wi-Fi 5): 5 GHz only, up to 3.5 Gbps, MU-MIMO. 802.11ax (Wi-Fi 6): 2.4, 5, and 6 GHz, up to 9.6 Gbps, OFDMA, better in crowded areas. 802.11be (Wi-Fi 7): latest, up to 46 Gbps.

Q183 What is the difference between 2.4 GHz and 5 GHz Wi-Fi?

Ans 2.4 GHz: Longer range (signal travels further and through walls better). Only 3 non-overlapping channels (1, 6, 11). More interference (microwave ovens, Bluetooth, and most older Wi-Fi all use 2.4 GHz). Slower maximum speeds. 5 GHz: Shorter range. Many non-overlapping channels (25+). Less congested — fewer devices use it. Much faster speeds. Rule of thumb: use 5 GHz when close to the AP for speed, 2.4 GHz for range and coverage through walls.

Q184 What is SSID?

Ans SSID (Service Set Identifier) is simply the name of a Wi-Fi network — what you see when you scan for networks on your phone. An access point broadcasts its SSID (by default) so users can find and connect. You can hide SSID (disable broadcast) for minor obscurity — but this is not real security because network scanners can still detect it. Multiple SSIDs can be configured on one physical AP — each mapped to a different VLAN.

Q185 What is the difference between autonomous AP and lightweight AP?

Ans Autonomous AP: standalone, self-contained. Each AP configured individually. Good for small deployments (1-3 APs) but unmanageable at scale (imagine configuring 500 APs one by one!). Lightweight AP (LWAP): thin device — all intelligence moved to a Wireless LAN Controller (WLC). AP just sends/receives radio signals; WLC handles authentication, roaming, configuration. All settings changed once on WLC = all APs updated instantly. Enterprise standard.

Q186 What is a Wireless LAN Controller (WLC)?

Ans A WLC is a device that centrally manages many lightweight APs. Functions: push configuration to all APs, manage client roaming (seamlessly moves client between APs), enforce security policies, radio frequency management (auto-adjust power and channels), monitor Wi-Fi health, handle authentication (802.1X with RADIUS). In Cisco environments: Cisco WLC 9800 series. This is CCNA Enterprise wireless architecture.

Q187 What is CAPWAP?

Ans CAPWAP (Control and Provisioning of Wireless Access Points) is the protocol used between Lightweight APs and the WLC. Two tunnels: Control tunnel (UDP 5246): carries management traffic, config, and commands — encrypted. Data tunnel (UDP 5247): carries client data traffic from AP to WLC. Why tunnel data? Centralised switching at WLC — all client traffic flows through WLC before reaching the rest of the network. This enables central policy enforcement.

Q188 What is local mode vs FlexConnect in wireless?

Ans Local mode (centralized): all client traffic tunnelled back to WLC via CAPWAP. WLC switches traffic to correct VLAN and network. Good when WLC is close (campus). FlexConnect mode: AP can locally switch traffic even when WAN link to WLC is down — clients keep internet access. Also locally switches traffic without sending it all the way to WLC (saves WAN bandwidth). Good for branch offices with limited WAN links.

Q189 What are the wireless LAN security protocols?

Ans WEP (Wired Equivalent Privacy): original, broken, never use — easily cracked in minutes. WPA (Wi-Fi Protected Access): temporary fix for WEP, uses TKIP — also considered weak now. WPA2: uses AES-CCMP encryption — strong, current standard. Two modes: Personal (pre-shared key password) and Enterprise (802.1X/RADIUS — unique credentials per user). WPA3: latest, fixes WPA2 weaknesses, mandatory for Wi-Fi 6 certified devices. Use WPA3 or WPA2-Enterprise.

Q190 What is TKIP vs AES in wireless security?

Ans TKIP (Temporal Key Integrity Protocol): replaced WEP's static key with a key that changes per-packet. Used in WPA. Now considered insecure. AES (Advanced Encryption Standard) with CCMP: much stronger block cipher encryption used in WPA2 and WPA3. AES is the US government encryption standard. Always use AES (WPA2 or WPA3) and avoid TKIP. In WPA2 config: always select AES, not TKIP mode.

Q191 What is wireless roaming?

Ans Wireless roaming is when a client device moves from one AP to another while maintaining its connection. In a well-designed network, roaming should be transparent — your phone call or video conference continues without interruption. Lightweight AP architecture (WLC) handles roaming automatically — the WLC knows which AP a client is associated with and can instantly redirect traffic. 802.11r, 802.11k, and 802.11v standards improve roaming speed and efficiency.

Q192 What are non-overlapping channels in 2.4 GHz Wi-Fi?

Ans The 2.4 GHz band has channels 1-13 (or 1-11 in USA), but each channel is 22 MHz wide and channels are only 5 MHz apart — they overlap heavily. Only channels 1, 6, and 11 do NOT overlap with each other. In a multi-AP deployment, use only channels 1, 6, and 11 to avoid interference between adjacent APs. Adjacent-channel interference is much worse than same-channel interference in 2.4 GHz.

Q193 What is wireless site survey?

Ans Before deploying wireless in a building, a site survey is done. Passive survey: walk the building with a laptop, measure signal strength from existing APs or planned locations. Active survey: connect to AP and measure actual throughput. Results: heat map showing signal coverage, dead spots, channel interference. This determines optimal AP placement and count. Without a site survey, you might install APs in the wrong locations and have poor coverage.

Q194 What is BSS and ESS?

Ans BSS (Basic Service Set): a single wireless cell — one AP and all its connected clients. Each BSS has a BSSID (the AP's MAC address). ESS (Extended Service Set): multiple BSSs with the same SSID, creating one large wireless network. Users can roam between APs in the ESS. IBSS (Independent BSS): ad-hoc mode — devices connect directly to each other without an AP. Home networks are BSS; corporate wireless is ESS.

Q195 What is RF interference and how does it affect Wi-Fi?

Ans RF interference disrupts wireless signals. Sources: Microwave ovens (2.4 GHz). Bluetooth devices (2.4 GHz). Neighbouring Wi-Fi networks using overlapping channels. Cordless phones (2.4 GHz). Physical obstacles: concrete walls, metal, water reduce signal strength. Effects: slow speeds, dropped connections, high retransmissions. Solutions: use 5 GHz (less interference), choose non-overlapping channels, conduct site survey, position APs away from interference sources.

Q196 What is MIMO?

Ans MIMO (Multiple Input Multiple Output) uses multiple antennas to send and receive multiple data streams simultaneously. Single antenna = one stream. 2x2 MIMO = two transmit, two receive antennas = better speed and reliability. 4x4 MIMO = four each. MU-MIMO (Multi-User MIMO): multiple streams to MULTIPLE clients simultaneously. Wi-Fi 5 supports 4x4 MU-MIMO. Wi-Fi 6 supports 8x8 MU-MIMO. More antennas = more capacity and speed in busy environments.

Q197 What is band steering?

Ans Band steering encourages dual-band capable devices (which support both 2.4 and 5 GHz) to connect on the 5 GHz band. Why? 5 GHz is faster and less congested. The WLC detects the device supports 5 GHz and either refuses the 2.4 GHz connection or sends a message recommending the 5 GHz network. Result: less congestion on 2.4 GHz (reserved for older devices that only support 2.4 GHz) and faster speeds on 5 GHz.

Q198 How do you troubleshoot Wi-Fi connectivity issues?

Ans Troubleshooting steps: 1) Can device see the SSID? (If not: AP down, wrong band, SSID hidden). 2) Authentication failing? (Wrong password, RADIUS server issue). 3) Getting IP address? (DHCP issue — getting 169.254.x.x?). 4) Can ping gateway? (IP connectivity). 5) Can access internet? (Routing issue). 6) Slow speeds? (Interference — use Wi-Fi analyser to check channel). 7) Drops frequently? (Low signal, high client count per AP).

Q199 What is the Cisco wireless management interface?

Ans The WLC management interface has an IP address for admin access — used by admins to configure and monitor the WLC via HTTPS or SSH. The WLC also has: AP Manager interface (for CAPWAP tunnels with APs), Virtual interface (for DHCP relay, web authentication redirect — always 1.1.1.1 or similar virtual IP), Dynamic interfaces (one per VLAN/SSID, connected to appropriate VLAN on the switch).

Q200 What is PSK (Pre-Shared Key) vs Enterprise wireless?

Ans PSK (Personal mode): one password shared by everyone. Simple to set up, suitable for home and small office. Weakness: if one user leaves (or password is compromised), you must change it for everyone. Enterprise mode (802.1X): each user has a unique username and password. Authenticated by RADIUS server (linked to Active Directory). Leaving employee's account disabled — others unaffected. No shared password. Mandatory for corporate security. Scale to thousands of users easily.

Q201 What is a rogue access point?

Ans A rogue AP is an unauthorised wireless access point connected to a company network — either by a malicious outsider or a well-meaning employee trying to extend Wi-Fi. Problems: bypasses all security, potentially bridging corporate LAN to an open wireless network. Detection: WLC scans for unknown APs, Wireless IDS alerts on new SSIDs, physical inspection. Prevention: 802.1X on all switch ports (rogue AP cannot get network access without authenticating).

Q202 What is WPA2 Personal four-way handshake attack?

Ans When a WPA2 Personal client connects, a 4-way handshake exchanges encrypted messages that are visible to anyone monitoring the wireless channel. An attacker captures this handshake, then runs an offline dictionary attack — trying millions of password guesses. If password is weak (common words, short), it can be cracked. WPA3 fixes this with SAE — the handshake cannot be cracked offline. Use strong, random passwords for WPA2 networks.

Q203 What is the difference between 802.11n and 802.11ac?

Ans 802.11n (Wi-Fi 4): works on both 2.4 GHz and 5 GHz, max channel width 40 MHz, up to 4 spatial streams, theoretical max ~600 Mbps. 802.11ac (Wi-Fi 5): 5 GHz ONLY, channel widths up to 160 MHz, up to 8 spatial streams, MU-MIMO downlink, theoretical max 3.5 Gbps. 802.11ac Wave 2 added MU-MIMO allowing simultaneous transmission to multiple clients. 802.11ac is the current standard in most enterprise APs.

Q204 What is OFDMA in Wi-Fi 6?

Ans OFDMA (Orthogonal Frequency Division Multiple Access) lets one AP transmit to multiple clients simultaneously by dividing a channel into smaller units called Resource Units (RUs). Previously, one transmission went to one client at a time (OFDM). Now one transmission serves 4, 8, or more clients simultaneously — each getting their slice of the channel. Result: much better efficiency in crowded environments (offices, hospitals, stadiums).

Q205 What is Target Wake Time (TWT)?

Ans TWT is a Wi-Fi 6 feature that lets the AP schedule exactly when IoT devices wake up to communicate, then go back to sleep. Without TWT, all IoT sensors must stay awake waiting for the channel to be available. With TWT, each sensor knows exactly when its turn is — it sleeps the rest of the time. Battery life of IoT sensors can increase from months to years. Essential for smart building and industrial IoT deployments.

Q206 What is a wireless guest network?

Ans A guest network is a separate SSID providing internet access to visitors without allowing them into the corporate network. Implemented with a separate VLAN: guest SSID maps to guest VLAN, which routes only to internet (firewall blocks access to corporate resources). Sometimes uses captive portal (web page requiring acceptance of terms before internet access). Keeps guests isolated from sensitive company systems.

Q207 What is the Cisco Unified Wireless Network architecture?

Ans Cisco's enterprise wireless solution: Lightweight APs connect to a WLC via CAPWAP. WLC connects to the LAN. Cisco DNA Center (management platform) provides centralised management, monitoring, and analytics for the entire network including wireless. Cisco ISE handles 802.1X authentication. APs use PoE from switches — no separate power needed. This architecture scales from small campus to thousands of APs globally.

Q208 What are wireless coverage and capacity planning considerations?

Ans Coverage planning: how far does each AP's signal reach? Depends on AP power, antenna gain, obstacles, and required signal strength (typically -65 to -70 dBm minimum). Capacity planning: how many clients per AP? Enterprise APs handle 20-50 clients well, struggle above 75. Dense deployments (lecture halls, offices) need more APs at lower power to create smaller cells. Client count > coverage distance for capacity-driven deployments.

Q209 What is wireless QoS (Wi-Fi Multimedia — WMM)?

Ans WMM (Wi-Fi Multimedia, based on 802.11e) provides QoS for wireless traffic. Four access categories (highest to lowest priority): Voice (VO), Video (VI), Best Effort (BE), Background (BK). Voice packets (VoIP) are transmitted with lower delay than background downloads. WMM is required for 802.11n and later. Without WMM, a large file download could ruin a wireless VoIP call by competing equally for airtime.

Q210 How do you secure a wireless network?

Ans

Wireless security checklist: Use WPA3 (or WPA2-AES at minimum). Use Enterprise mode (802.1X) for corporate networks. Disable WEP and TKIP. Change default admin credentials on WAP/WLC. Disable SSID broadcast (minor — not real security). Enable rogue AP detection. Use separate guest VLAN for visitors. Implement wireless IDS/IPS. Ensure firmware is updated. Use strong, unique passwords. Enable MFP (Management Frame Protection).

SECTION 08

Network Automation & Programmability

Q211 –
Q240

SDN · REST API · Ansible · Python · JSON · NETCONF · DNA Center · DevNet

Q211 What is network automation and why is it important?

Ans Network automation uses software to configure, manage, and monitor network devices automatically — instead of logging into each device and typing commands manually. Why it matters: a network with 500 switches takes weeks to configure manually, seconds with automation. Benefits: eliminates human typing errors, consistent configuration across all devices, faster deployments, automatic rollback if something goes wrong. CCNA now includes automation basics.

TIP *Network automation is 10% of CCNA exam. Understand concepts well — not deep coding skills!*

Q212 What is SDN (Software Defined Networking)?

Ans SDN separates the control plane (the brain that decides where to send traffic) from the data plane (the hands that actually forward packets). Traditional: each router/switch makes its own decisions. SDN: a centralised controller makes decisions for all devices. Benefits: entire network programmed from one place, no need to log into individual devices, faster policy changes. Cisco DNA Center is Cisco's SDN controller.

Q213 What is the difference between the data plane, control plane, and management plane?

Ans Data plane (Forwarding plane): actually forwards packets based on routing/switching tables — this happens in hardware at wire speed. Control plane: builds and maintains routing tables — routing protocols (OSPF, EIGRP) run here. Management plane: used by network admins — SSH, Telnet, SNMP, Syslog. In SDN, the control plane is removed from individual devices and centralised in a controller, while the data plane remains on the devices.

Q214 What is a northbound and southbound API in SDN?

Ans In SDN architecture: Southbound API: communication between the SDN controller and the actual network devices (switches/routers). Examples: OpenFlow, NETCONF, RESTCONF. The controller uses these to program the devices. Northbound API: communication between the SDN controller and applications or management tools above it. Examples: REST API. Applications use northbound APIs to tell the controller what they need.

Q215 What is a REST API?

Ans REST API (Representational State Transfer) is the most common way applications communicate with network controllers. It uses HTTP/HTTPS with four main operations: GET (retrieve data — read-only), POST (create something new), PUT (update existing data), DELETE (remove something). Data is sent and received in JSON or XML format. Example: your monitoring tool sends a GET request to Cisco DNA Center API to retrieve list of all network devices.

Q216 What is JSON and why is it used in networking?

Ans JSON (JavaScript Object Notation) is a lightweight, human-readable data format used to exchange data between systems. Example: `{'hostname': 'Router1', 'ip': '192.168.1.1', 'status': 'up'}`. Network automation tools use JSON to send configurations to devices and receive status information back. It is much easier for software to read JSON than trying to parse the text output of 'show' commands.

Q217 What is YAML?

Ans YAML (Yet Another Markup Language) is a human-readable data format often used in configuration files for automation tools like Ansible. It is even more readable than JSON — uses indentation instead of brackets. Example: `hostname: Router1, interfaces: - name: G0/0, ip: 192.168.1.1`. Ansible playbooks are written in YAML. Many network automation scripts use YAML files to define device configurations that are then pushed to devices.

Q218 What is Ansible and how is it used in networking?

Ans Ansible is an automation tool that configures network devices from a central control machine — no software installed on the devices. You write Playbooks (YAML files describing what to configure). Ansible connects to devices via SSH and runs the commands. Benefits: same playbook configures 1 or 1000 devices in minutes. Idempotent: running the same playbook twice does not break anything — it only makes changes if needed. Great for Cisco IOS devices.

Q219 What is Python in networking context?

Ans Python is the most popular programming language for network automation. Network engineers use Python to: automatically generate configurations from templates, retrieve data from devices and process it (using Netmiko, NAPALM libraries), interact with REST APIs (using Requests library), automate repetitive tasks (backup configs, check interfaces). You do not need to be an expert programmer — basic Python (variables, loops, functions) is enough for CCNA level.

Q220 What is Cisco DNA Center?

Ans Cisco DNA Center (now Cisco Catalyst Center) is Cisco's network management and automation platform. Features: Design (draw network topology, define IP plans), Policy (create security and QoS policies applied across the network), Provision (deploy configs to devices automatically), Assurance (AI-powered monitoring and issue detection), API (REST API for custom integrations). One platform manages the entire campus network — switches, routers, and wireless.

Q221 What is Cisco DevNet?

Ans Cisco DevNet (developer.cisco.com) is Cisco's developer community providing APIs, SDKs, documentation, and sandbox environments to learn and build network automation. Free sandboxes: you can practice with real Cisco DNA Center, ACI, and IOS XE APIs without owning any hardware. CCNA exam covers DevNet concepts. The DevNet Associate certification is Cisco's dedicated automation certification — a great next step after CCNA.

Q222 What is NETCONF?

Ans NETCONF (Network Configuration Protocol) is a standardised protocol for configuring network devices programmatically. Uses XML data and SSH transport. Operations: get (retrieve config/state), edit-config (change configuration), copy-config, delete-config, lock (prevent others from changing config while you work). Cisco IOS XE supports NETCONF. Combined with YANG data models, it is the foundation of modern network automation.

Q223 What is RESTCONF?

Ans RESTCONF is a newer, simpler alternative to NETCONF that uses REST API (HTTP/HTTPS) instead of SSH+XML. Uses JSON or XML. Same YANG data models as NETCONF. Easier to use (any tool that can send HTTP requests works) but slightly less powerful than NETCONF. Enable on Cisco IOS XE: `restconf` `management-plane` `access-list` allowing management hosts. Access via HTTPS: `GET https://router-ip/restconf/data/...`

Q224 What is a YANG model?

Ans YANG (Yet Another Next Generation) is a data modelling language that defines the structure of configuration and operational data for network devices. Think of it as a schema that says: a router has interfaces, each interface has a name, IP address, and enabled/disabled state — in a structured, machine-readable way. NETCONF and RESTCONF use YANG models to know what data a device supports and how it should be structured.

Q225 What is the difference between imperative and declarative automation?

Ans Imperative: you specify exactly WHAT to do and HOW to do it step-by-step — 'Log in to router, go to interface config, type this IP address command.' Traditional CLI scripting is imperative. Declarative: you specify the desired END STATE — 'I want interface G0/0 to have IP 192.168.1.1' — and the tool figures out how to achieve it. Ansible is declarative. DNA Center is declarative. Declarative is safer — no risk of doing wrong steps.

Q226 What is Cisco IOS XE?

Ans IOS XE is Cisco's modern operating system for enterprise routers and switches (Catalyst 9000 series, ASR routers). It runs IOS as a process on top of a Linux kernel — more modular and maintainable than classic IOS. Benefits: supports Python scripting on-device, RESTCONF/NETCONF, advanced automation features, in-service software upgrades (no reboot during patching). IOS XE is the future of Cisco enterprise networking.

Q227 What is the Cisco DevNet sandbox?

Ans DevNet sandboxes are free virtual lab environments Cisco provides at developer.cisco.com. Types: Always-On sandboxes (instantly accessible, shared, limited control), Reservation sandboxes (book for a time slot, dedicated access, full control). Available environments: IOS XE routers, DNA Center, ACI fabric, SD-WAN, Security. Perfect for learning REST API calls, automation, and testing scripts without any hardware. Free to use with a Cisco account.

Q228 What is Git and why do network engineers use it?

Ans Git is a version control system that tracks changes to files over time. Network engineers use it to: store device configurations (like code), track what changed and when, revert to previous working config if something breaks, collaborate (multiple team members working on config files), backup all configurations in a repository (GitHub, GitLab). 'Infrastructure as Code' — treat network configs like software code.

Q229 What is a configuration template?

Ans A configuration template is a skeleton configuration file with variables that get filled in for each device. Example: a template for switch configuration with variables for hostname, VLAN IDs, management IP, and port assignments. When deploying 50 branch switches, fill in the variables for each and generate 50 unique configs automatically. Tools: Cisco DNA Center templates, Ansible with Jinja2 templates. Eliminates copy-paste errors and ensures consistency.

Q230 What is network telemetry?

Ans Network telemetry is the automatic collection of performance data from network devices in real time. Traditional monitoring: SNMP polls devices every 5 minutes (can miss issues). Telemetry (streaming telemetry): devices continuously push data to a collector at high frequency — see problems as they happen. Data types: interface counters, CPU/memory, routing table changes, environmental sensors. Cisco DNA Center and modern platforms support telemetry.

Q231 What is the difference between on-box and off-box automation?

Ans On-box automation runs automation scripts directly ON the network device. Cisco IOS XE supports Python scripts running on the router itself — triggered by events (a port goes down, CPU spikes). Useful for rapid local response without needing a controller. Off-box automation runs on a separate server that connects to devices via SSH or API. Ansible, Python with Netmiko — more common, centralised, easier to manage at scale.

Q232 What is SNMP polling vs streaming telemetry?

Ans SNMP polling: management station asks device for data every X minutes. Simple, widely supported, but can miss short-duration problems. Streaming telemetry: device proactively pushes data to collector at high frequency (every second or less). Detects brief spikes in CPU or bandwidth that SNMP would miss. Uses YANG models to define what data is pushed. Protocols: gRPC, gNMI, NETCONF subscriptions. Modern networks prefer streaming telemetry for real-time visibility.

Q233 What is infrastructure as code (IaC)?

Ans Infrastructure as Code means defining and managing network/server infrastructure using configuration files (code) instead of manual processes. Benefits: entire infrastructure in Git (version control), deploy or rebuild in minutes, guaranteed consistency (same code = same result every time), peer review before applying changes. For networking: Ansible playbooks, Terraform, Python scripts define the desired network state. IaC is standard practice in modern enterprise IT.

Q234 What is the model-driven programmability architecture?

Ans Model-driven programmability uses YANG data models to provide a structured, machine-readable interface to network devices. Instead of parsing text output of show commands, tools can request structured data. Architecture: YANG model defines data structure → NETCONF or RESTCONF transports it → Controller or automation tool processes it. This enables reliable, vendor-neutral automation. Cisco IOS XE, NX-OS, and IOS XR all support model-driven programmability.

Q235 What is Cisco Meraki?

Ans Cisco Meraki is a cloud-managed networking platform. All Meraki devices (switches, access points, firewalls) are managed entirely through a web dashboard — no CLI configuration. Configuration changes are pushed from the cloud. Benefits: simple management, no on-site configuration, automatic updates, centralised visibility. Best for: distributed small offices, retail chains, schools. CCNA covers Meraki as an example of cloud-managed versus on-premises managed networks.

Q236 What is the Cisco Catalyst Center (DNA Center) REST API?

Ans DNA Center provides a comprehensive REST API allowing integration with other tools. Example API calls: GET /intent/api/v1/network-device (get list of all network devices), POST /dna/intent/api/v1/template-programmer/project (create a template), GET /dna/intent/api/v1/issues (get current network issues). Authentication: first POST to /api/system/v1/auth/token to get a session token, then include token in all subsequent requests.

Q237 What is event-driven automation?

Ans Event-driven automation triggers an automated response when a specific event happens — no human needed. Examples: Interface goes down □ automation script sends Slack alert and attempts to bring it back up. New employee added to HR system □ automation provisions their network access. CPU exceeds 90% □ automation generates a ticket and collects diagnostic data. Cisco IOS XE supports EEM (Embedded Event Manager) for on-box event-driven automation.

Q238 What is ChatOps?

Ans ChatOps integrates IT operations with chat platforms (Slack, Microsoft Teams). Network engineers type commands in a chat channel, a bot executes them on network devices, and results appear in the chat. Benefits: entire team sees what actions were taken (transparency), natural audit trail, easy collaboration. Example: type '@netbot show interface status Router1' in Slack □ bot runs the command and posts results. Cisco and open-source tools (Nautobot, Errbot) support ChatOps.

Q239 What is Postman and how is it used in networking?

Ans Postman is a popular tool for testing and working with REST APIs. You can: send GET/POST/PUT/DELETE requests to any API, view responses in formatted JSON, save and organise API requests into collections, set variables for common values (server IP, authentication token). Network engineers use Postman to explore Cisco DNA Center, Meraki, and other vendor APIs before writing automation scripts. Free to download and use.

Q240 What is the future of network engineering?**Ans**

Network engineering is shifting from manual CLI configuration to automated, intent-based networking. Future skills needed: Python scripting and REST API knowledge, understanding of automation platforms (Ansible, Terraform), cloud networking (AWS VPC, Azure networking), SD-WAN and SASE, cybersecurity integration. Traditional CLI skills remain important for troubleshooting, but automation handles routine changes. CCNA + CCNP + DevNet = complete modern network engineer skillset.

SECTION 09

WAN & Network Troubleshooting

Q241 –
Q270

WAN Technologies · VPN · Troubleshooting Methodology · CLI Diagnostics

Q241 What is a WAN and how does it differ from a LAN?

Ans WAN (Wide Area Network) connects geographically separate locations — different buildings, cities, or countries. Unlike LAN (which you own and control), WAN links are usually rented from a service provider (telecom company). WAN characteristics: lower bandwidth than LAN, higher cost, higher latency, shared infrastructure. Your branch office connects to HQ via WAN. The internet is the world's largest public WAN.

TIP CCNA covers WAN basics — know technology types and when each is appropriate!

Q242 What are the main WAN connectivity types?

Ans Leased line (T1/E1): dedicated point-to-point copper or fibre, constant bandwidth, expensive. MPLS (Multiprotocol Label Switching): provider-managed network, any-to-any connectivity, supports QoS, SLA-backed — most used in enterprise. Metro Ethernet: carrier Ethernet service, familiar interface, scalable bandwidth. DSL: broadband over telephone lines, good for small offices. Cable: DOCSIS over coax, shared bandwidth. Fibre (FTTH): best performance. 4G/5G cellular: mobile WAN or backup. Satellite: remote areas.

Q243 What is MPLS and why do enterprises use it?

Ans MPLS (Multiprotocol Label Switching) is a provider WAN service. Instead of routing each packet by IP, MPLS labels packets and switches them along pre-defined Label Switched Paths (LSPs) — much faster. Why enterprises love it: connect all offices in any-to-any topology (branch can talk directly to branch, not everything through HQ), guaranteed QoS (voice and video get priority), SLA with guaranteed uptime. Downside: expensive. SD-WAN is increasingly replacing MPLS.

Q244 What is SD-WAN and why is it replacing MPLS?

Ans SD-WAN (Software-Defined WAN) manages multiple WAN connections (broadband, MPLS, LTE) intelligently in software. A cheap 100 Mbps broadband link can carry most traffic. MPLS kept for voice/video. If broadband fails, SD-WAN instantly switches traffic to MPLS or LTE. Benefits: 50-80% cost reduction vs pure MPLS, application-aware routing (automatically uses best link per application), centralised management. That is why SD-WAN adoption is exploding.

Q245 What is a site-to-site VPN?

Ans A site-to-site VPN connects two office networks securely over the internet using IPsec encryption. Both ends have a router/firewall that creates the tunnel. Once configured, all users at both sites can communicate as if directly connected — no client software needed on their PCs. Cost advantage: cheap internet connection + VPN replaces expensive dedicated WAN link. Common for small businesses that cannot afford MPLS.

Q246 What is a remote access VPN?

Ans A remote access VPN allows individual users (employees working from home or travelling) to securely connect to the company network over the internet. User installs VPN client software (Cisco AnyConnect). Client connects to company VPN gateway (router or firewall). Encrypted tunnel created — user's traffic goes through tunnel to company network. User accesses internal resources as if sitting in the office. Essential for remote work.

Q247 What is GRE tunnel and when is it used?

Ans GRE (Generic Routing Encapsulation) wraps one packet inside another — creates a virtual point-to-point tunnel over any IP network. Common uses: carry routing protocol updates over the internet between sites (OSPF or EIGRP cannot run natively over internet), connect islands of private address space, transport multicast over non-multicast networks. GRE has no encryption — combine with IPsec (GRE over IPsec) for security.

Q248 What is the CompTIA/Cisco troubleshooting methodology?

Ans Standard troubleshooting process: 1) Define the problem (what exactly is not working? When did it start?). 2) Gather information (show commands, logs, user reports). 3) Consider possible causes (eliminate the obvious first). 4) Create action plan (what will you try and in what order?). 5) Implement solution (one change at a time). 6) Verify the problem is fixed. 7) Document what you did (so others can learn). Never make multiple changes simultaneously — impossible to know what fixed it.

TIP

Methodical troubleshooting impresses interviewers far more than random guessing!

Q249 What is the OSI model troubleshooting approach?

Ans Troubleshoot from the bottom up (most common): Layer 1 (Physical): is cable plugged in? Link lights? Try different cable/port. Layer 2 (Data Link): correct VLAN? STP blocking? 'show interfaces', 'show vlan'. Layer 3 (Network): correct IP/subnet/gateway? Ping default gateway. Layer 4 (Transport): TCP session establishing? ACL blocking the port? Layer 7 (Application): correct server IP, port open, service running? This systematic approach quickly isolates which layer has the problem.

Q250 What does 'ping' output tell you on a Cisco router?

Ans !!!!!(5 of 5 success): connectivity is working.(0 of 5): complete failure — check routing, ACLs, interface status. ...!!(2 of 5 failed): first two failed (ARP resolution delay), then worked — normal first-time behavior. U (Destination Unreachable): router has no route to destination or destination returned unreachable ICMP message. ? (Unknown packet): firewall dropping and returning nothing. Each character represents one packet's fate.

Q251 How do you troubleshoot 'interface is down/down'?

Ans Interface down/down (both Line Status and Protocol down): Physical issue — no cable connected, cable broken, wrong cable type (fibre vs copper), far-end device off. Steps: check cable is plugged in both ends, try a known-good cable, verify far-end device is on, check speed/duplex settings match both ends. 'Administratively down' means 'shutdown' was applied — use 'no shutdown' to bring it up. Protocol down with Line up = Layer 2 issue.

Q252 How do you troubleshoot a PC that cannot reach the internet?

Ans Systematic approach: 1) ping 127.0.0.1 (is TCP/IP working?). 2) ping own IP (is NIC working?). 3) ping default gateway (is LAN working?). 4) ping 8.8.8.8 (is internet IP reachable?). 5) ping google.com (is DNS working?). If step 3 fails: check IP address, subnet mask, gateway settings. If step 4 fails: router/firewall issue, ISP issue. If step 5 fails but step 4 works: DNS server unreachable or misconfigured.

Q253 What is 'debug' command and when should you use it?

Ans Debug commands show real-time messages as events happen on the router — extremely detailed. Examples: debug ip ospf events (watch OSPF adjacency forming), debug ip dhcp server events (see DHCP exchanges). Warning: debug generates massive output on a busy router and can crash it by consuming all CPU. Always limit debug: debug ip ospf adj (not all OSPF). Always disable when done: undebug all or 'no debug all'. Use only in lab or low-traffic maintenance windows.

Q254 What is a CDP neighbour table and how does it help troubleshooting?

Ans Show cdp neighbors detail shows all directly connected Cisco devices including their hostname, IP address, device type, IOS version, and interface. In troubleshooting: 'show cdp neighbors' on each switch helps you quickly map physical connections — which port connects to which switch. Particularly valuable when physical documentation is outdated or missing. Can also reveal if interfaces are connected to unexpected devices.

Q255 What are common OSPF troubleshooting issues?

Ans OSPF neighbours not forming? Check: both routers in same area? Same subnet? (show interfaces). Hello and dead timers match? (show ip ospf interface). Same authentication configured? MTU mismatch? (show interfaces — MTU should match). Network statement correct? (show ip protocols). Stub area mismatch? Routes not appearing? Check: network statement correct, routes being redistributed, ACL not blocking OSPF traffic (protocol 89).

Q256 What are common VLAN troubleshooting issues?

Ans VLAN troubleshooting: PC in VLAN 10 cannot reach server in VLAN 10 on different switch: Is VLAN 10 trunk-allowed between switches? (show interfaces trunk). Is the inter-switch link actually a trunk? PC in VLAN 10 cannot reach server in VLAN 20: Is there a Layer 3 device routing between VLANs? Does SVI exist for both VLANs? Are SVIs up? Is IP routing enabled on L3 switch? (ip routing). Is default gateway on PC pointing to correct SVI?

Q257 What is 'debug ip packet' and its risks?

Ans Debug ip packet shows every IP packet processed by the router CPU. This includes source/destination IP and protocols. Very useful for seeing if packets are arriving and where they are going. MAJOR RISK: on a production router handling millions of packets per second, this WILL overwhelm the CPU and potentially crash the router, causing an outage. Only use with ACL to limit to specific IPs: debug ip packet 101 detail (where ACL 101 matches only the specific traffic you need to debug).

Q258 What is the 'show processes cpu' command?

Ans Show processes cpu displays current CPU utilisation and the top processes consuming CPU. Output shows: 5-second average, 1-minute average, 5-minute average CPU utilisation. Below each: list of processes with their % CPU. High CPU causes: routing protocol flooding, DDoS attack, software bug, large routing table recalculation. If OSPF is consuming high CPU: check for topology instability (interface flapping). Schedule: show processes cpu history (graph of last hour).

Q259 How do you verify NAT is working correctly?

Ans Key verification: show ip nat translations (see active translation table — should show inside local IP mapped to inside global/public IP), show ip nat statistics (total translations, hits, misses, expired). Troubleshooting: if no translations appear, check ACL matches traffic correctly, inside/outside interfaces correctly marked, NAT statement references correct ACL and interface. Debug: debug ip nat (shows each translation event — use carefully).

Q260 What is the 'logging buffered' command?

Ans Logging buffered [size] stores log messages in router RAM buffer — viewable with 'show logging'. Default buffer size is small. Increase for more log history: logging buffered 65536. Without this, log messages display on console but are not stored. Important for troubleshooting: 'show logging' shows recent events with timestamps (time synchronization via NTP is critical for accurate timestamps). Severity level filter: logging buffered 6 (up to informational level).

Q261 What is a traceroute and what does * * * mean?

Ans Traceroute sends probes with increasing TTL to reveal the path to a destination. '* * *' means a router did not respond — either it has ICMP disabled (drop silently) or there is a routing problem. Key insight: if you see success up to hop N then * * * — the problem is at or after hop N's next router. If hop 1 shows * * * — your default gateway is unreachable or blocking ICMP. Three asterisks per hop = three individual probe packets sent per TTL value.

Q262 What is a broadcast storm and how do you detect it?

Ans Broadcast storm: massive amounts of broadcast traffic flooding the network, consuming all bandwidth, crashing switches and routers. Cause: a Layer 2 loop (STP not running or STP failure). Detection: all switch interfaces showing 100% utilisation, unable to SSH to switches (CPU overwhelmed), users report complete network outage, interface counters rapidly increasing for broadcast/multicast. Fix: disconnect redundant links until loop is identified, verify STP is running on all switches.

Q263 What is interface error counter analysis?

Ans Switch/router interface counters reveal problems: Input errors / CRC errors: bad cable, NIC failing, duplex mismatch — causing corrupt frames. Giants: frames larger than 1518 bytes — VLAN tagging issue or MTU problem. Runts: frames smaller than 64 bytes — collision or NIC issue. Collisions: should be ZERO on full-duplex — if non-zero, duplex mismatch. Output drops: queue overflow — interface congested, need QoS or more bandwidth. View: show interfaces GigabitEthernet 0/0.

Q264 What is Wireshark and how is it used?

Ans Wireshark is a free packet capture and analysis tool. It captures all packets on a network interface and displays them with full details. Network engineers use it to: see if packets are arriving at all, verify correct IP/MAC addresses, analyse TCP handshakes, detect ARP spoofing, find rogue DHCP servers, troubleshoot application protocols. In a switched environment, you need to configure a SPAN (mirror) port on the switch to capture traffic from other ports.

Q265 How do you check interface speed and duplex on Cisco?

Ans Command: show interfaces GigabitEthernet 0/0. Look for line like: 'Full-duplex, 1000Mb/s' or 'Half-duplex, 100Mb/s'. Duplex mismatch symptoms: interface showing input errors, very slow throughput, intermittent connectivity. Fix: explicitly set speed and duplex on both ends: interface g0/0 □ duplex full □ speed 1000. Or set both ends to auto and allow auto-negotiation. Never mix one end manual and one end auto — this always causes mismatch.

Q266 What is split horizon and how does it prevent loops?

Ans Split horizon is a routing rule: do not advertise a route back out the interface you learned it from. Why it prevents loops: Router A learns about Network X from Router B via interface 0/0. Split horizon prevents Router A from advertising Network X back to Router B on that same interface. Without split horizon, Router B might learn from Router A that Router A can reach Network X via Router B — circular!

Q267 What is the 'no ip routing' command and what does it do?

Ans By default, Layer 3 switches have ip routing disabled — they act as pure Layer 2 switches. The 'ip routing' command enables the switch to route between VLANs using SVIs. The 'no ip routing' command disables this. If you notice a Layer 3 switch is not routing between VLANs despite correct SVI configuration, check: is 'ip routing' in the running config? If not, that is your problem. Add it and routing works immediately.

Q268 How do you troubleshoot DHCP issues?

Ans Client not getting IP address (getting 169.254.x.x or nothing): 1) Is DHCP server running? (show ip dhcp pool). 2) Is there an active scope? (show ip dhcp binding). 3) Is the pool exhausted? (show ip dhcp pool — look at In Use count). 4) Excluded addresses too broad? 5) DHCP relay configured if server on different subnet? (ip helper-address). 6) ACL blocking UDP 67/68? Test: from router, 'debug ip dhcp server events'.

Q269 What is the purpose of 'ip route 0.0.0.0 0.0.0.0 Null0'?

Ans This route sends all unmatched traffic to Null0 — a virtual interface that silently discards packets. Used with BGP or route summarisation to prevent routing loops. Example: Router summarises 10.0.0.0/8 to a neighbour. Without a Null0 route, if a more specific sub-network within 10.0.0.0/8 is not in the routing table, the default route might forward it somewhere wrong, causing a loop. The Null0 route acts as a safety net.

Q270 What is a debug vs show command and when do you use each?

Ans Show commands: read-only, display current state (routing table, interfaces, configs) — safe to run anytime, no impact on device. Use for: regular monitoring, verification, troubleshooting. Debug commands: real-time event messages, show what is happening as it happens — can consume significant CPU. Use for: specific problems that show commands cannot diagnose, in low-traffic windows or lab, always disable when done (undebug all). Rule: always try show commands first.

SECTION 10	CCNA Career & Exam Preparation	Q271 – Q300
Exam Tips · Career Paths · CCNP · Salary · Study Strategy · Lab Practice		
Q271	What is the CCNA exam (200-301) and what does it cover?	
Ans	CCNA 200-301 is Cisco's entry-level networking certification — one exam covering everything: Network Fundamentals (20%), Network Access - switching/wireless (20%), IP Connectivity - routing (25%), IP Services - DHCP, NAT, QoS (10%), Security Fundamentals (15%), Automation and Programmability (10%). The exam is approximately 120 questions, 120 minutes. Passing score: around 825/1000 (exact pass mark not published). Exam available at Pearson VUE testing centres.	
TIP	<i>Study each domain proportionally — IP Connectivity (routing) is the largest domain at 25%!</i>	
Q272	What is the best study strategy for CCNA?	
Ans	Proven strategy: 1) Use official Cisco study guide (OCG by Wendell Odom — considered the bible of CCNA). 2) Watch video course for visual learning (Jeremy's IT Lab on YouTube — free and excellent). 3) Practice labs DAILY in Cisco Packet Tracer (free). 4) Take practice exams when you think you are ready. 5) Review wrong answers deeply — understand WHY, not just memorise. 6) Focus most time on routing (OSPF), switching (VLANs, STP), and security. Aim for 85%+ on practice exams before booking the real test.	
Q273	What is Cisco Packet Tracer and how do you get it?	
Ans	Cisco Packet Tracer is a free network simulation software by Cisco. Download: create a free account at netacad.cisco.com □ enroll in free 'Getting Started with Cisco Packet Tracer' course □ download included. It simulates routers, switches, wireless access points, PCs, servers, and IoT devices. Practice all CCNA commands without needing real hardware. Very accurate for most CCNA topics. GNS3 is an alternative that runs real IOS images.	
Q274	What career opportunities does CCNA open?	
Ans	CCNA-certified roles: Network Administrator, Network Engineer (Junior), IT Support Engineer, NOC Engineer, System Administrator. Salary ranges (India): Entry level with CCNA: ₹4-8 LPA, Mid-level (2-3 years experience): ₹8-15 LPA, Senior Network Engineer (5+ years): ₹15-25 LPA. CCNA also highly valued for: government sector IT, defence networks, banking IT infrastructure, cloud service providers.	
TIP	<i>CCNA + real hands-on experience + CCNP = strong career foundation in networking!</i>	

Q275 What is CCNP and how does it build on CCNA?

Ans CCNP (Cisco Certified Network Professional) is the next level after CCNA. Specialisations: CCNP Enterprise (routing, switching, SD-WAN, wireless), CCNP Security (firewalls, VPN, identity), CCNP Data Center, CCNP Service Provider. CCNP requires: one core exam + one concentration exam. Much deeper and more technical than CCNA. With CCNA + CCNP Enterprise: senior network engineer roles, salaries ₹15-30 LPA in India, \$80K-\$120K+ in USA.

Q276 What is CCIE and how hard is it to achieve?

Ans CCIE (Cisco Certified Internetwork Expert) is the most prestigious networking certification in the world. Requires: written qualification exam + 8-hour practical lab exam at Cisco testing centres globally. Pass rate: approximately 25-30% on first attempt. Requires 3-5 years of deep hands-on experience. CCIE holders are rare and highly paid: India ₹30-60 LPA+, USA \$120K-\$200K+. Current tracks: Enterprise Infrastructure, Security, Data Centre, Service Provider, Wireless, Collaboration.

Q277 What is Cisco NetAcad?

Ans Cisco Networking Academy (NetAcad.com) is Cisco's free and paid online learning platform. Key courses: Introduction to Networks (ITN), Switching Routing and Wireless Essentials (SRWE), Enterprise Networking Security and Automation (ENSA) — together these three courses cover the full CCNA curriculum. Free enrollment options available. Labs included. Used by millions of students and professionals globally. Also offers Cybersecurity, Python, IoT, and other courses.

Q278 What type of questions appear on the CCNA exam?

Ans Question types: Multiple choice (single best answer — most common), Multiple select (choose 2 or 3 correct answers), Drag and drop (match items — e.g. match protocol to port number), Configuration simulation (limited — type commands in simulated CLI), Testlet (scenario with multiple questions based on same topology). No open-ended questions. Time management: 120 minutes for approximately 120 questions = 1 minute per question average.

Q279 How do you use Cisco Packet Tracer for CCNA lab practice?

Ans Effective Packet Tracer practice: Build basic topologies (2 routers + 1 switch + 2 PCs). Configure each concept as you learn it: basic routing, OSPF, VLANs, STP, NAT, DHCP. Use simulation mode to see packets travelling between devices (visual understanding). Practice troubleshooting: break a working topology then fix it. Use pre-built CCNA labs from the internet. Jeremy's IT Lab (YouTube) includes free Packet Tracer files matching his videos.

Q280 What is the difference between CCNA and CompTIA Network+?

Ans CompTIA Network+: vendor-neutral, covers all networking concepts broadly, no vendor-specific configuration. Good general foundation. Cisco CCNA: Cisco-specific, includes deep configuration skills (Cisco IOS commands), more respected in enterprise environments. CCNA shows you can actually configure networks, not just explain them. Recommendation: if aiming for Cisco network engineering roles, go for CCNA. Network+ is good alongside CCNA for broader IT generalist knowledge.

Q281 What are subnetting quick tips for the CCNA exam?

Ans Fast subnetting shortcuts: Remember block sizes: /24=256, /25=128, /26=64, /27=32, /28=16, /29=8, /30=4. Hosts = block size - 2. Network address = first IP in block, Broadcast = last IP in block, Usable = everything in between. For /27 containing 192.168.1.50: block of 32 starting at 0, 32, 64, 96... 50 falls in 32-63 block. Network=192.168.1.32, Broadcast=192.168.1.63, Gateway usually .33 or .62. Practice 20 subnets daily.

Q282 What are the most common CCNA exam topics to focus on?

Ans Highest-priority topics: OSPF (single-area, neighbour states, DR/BDR, cost), VLANs and trunking (802.1Q, access vs trunk ports, inter-VLAN routing), STP/RSTP (root bridge election, port roles, PortFast, BPDU Guard), IPv4 subnetting (master this — appears in many questions), NAT (PAT configuration and verification), ACLs (standard and extended, wildcard masks), DHCP, SSH configuration, basic wireless concepts.

Q283 What is a practical example of a CCNA configuration task?

Ans Common exam scenario: Configure Router1 and Router2 connected via serial link. Router1 LAN: 192.168.1.0/24. Router2 LAN: 192.168.2.0/24. WAN: 10.0.0.0/30. Tasks: Assign IPs to all interfaces (no shutdown), configure OSPF area 0 on both routers (network statements), verify neighbours form (show ip ospf neighbor), verify all networks in routing table (show ip route), test connectivity PC1 to PC2 (ping across both routers). This covers interfaces, routing, and verification in one exercise.

Q284 How do you approach a CCNA practice exam?

Ans Strategy: Read every question TWICE before answering. Eliminate obviously wrong answers first. For configuration questions: mentally trace through what the commands would do. For troubleshooting scenarios: apply OSI model bottom-up approach. Mark unsure questions and return. For multiple-select: each answer is either definitely right, definitely wrong, or maybe — start with what you are sure about. Time: if stuck for more than 2 minutes, mark and move on.

Q285 What are common CCNA exam mistakes to avoid?

Ans Common traps: Confusing AD and metric (AD selects which PROTOCOL's route to use; metric selects best path WITHIN a protocol). Confusing subnet mask and wildcard mask (they are inverses). Thinking 'no shutdown' is automatic (it is NOT on router interfaces). Forgetting that STP convergence takes 30-50 seconds (not instant). Confusing trunk 'allowed VLAN' (which VLANs can pass) with 'native VLAN' (untagged VLAN). Mixing up OSPF cost direction (lower cost = preferred, opposite of hop count).

Q286 What labs should you build to prepare for CCNA?

Ans Essential practice labs: Basic router config (hostname, passwords, SSH, interfaces). VLAN and inter-VLAN routing (L3 switch with SVIs). OSPF single-area (3 routers, verify full adjacency). Static routing (with default route to simulated internet). NAT overload (PAT — inside to outside). DHCP server on router. ACL (permit/deny specific traffic). STP (add redundant links, verify root bridge, test PortFast). Wireless basics (WLC config if Packet Tracer supports it).

Q287 What is the CCNA exam recertification policy?

Ans CCNA certification is valid for 3 years. To recertify: pass current CCNA exam again, pass a CCNP core exam (which automatically recertifies CCNA), pass CCIE written exam, or earn Cisco Continuing Education (CE) credits (50 CE credits recertifies CCNA). Cisco recommends staying current with technology — the recertification cycle encourages updated knowledge. Cisco tracks certifications at certificationtracker.cisco.com.

Q288 What is the importance of hands-on lab practice?

Ans The CCNA exam tests your ability to configure and troubleshoot — not just recall facts. Reading about OSPF is different from configuring it, watching it form neighbours, and fixing it when it does not. Hand-on practice: builds real muscle memory for commands, makes troubleshooting instinctive, reveals exactly which concepts you have gaps in (you think you understand VLANs until you cannot make them work!), and prepares you for real work after the exam. No shortcut to lab time.

Q289 What soft skills complement technical CCNA skills?

Ans Technical skills get you hired; soft skills get you promoted: Clear communication (explain network issues to non-technical management), Documentation discipline (maintain accurate network diagrams and change logs), Problem-solving methodology (structured, calm approach under pressure), Teamwork (collaborate with server/application/security teams), Customer service mindset (internal users are your customers), Continuous learning (technology changes — those who keep learning stay relevant).

Q290 What is Cisco's learning platform and certifications structure?

Ans Cisco certification tracks and levels: Entry (CCT), Associate (CCNA), Professional (CCNP), Expert (CCIE), Architect (CCAr). Specialisations: Enterprise, Security, Data Centre, Service Provider, Wireless, Collaboration, Cyber Ops, DevNet. Cisco Learning Network (learningnetwork.cisco.com): free study materials, labs, and community. Cisco Press books are the official study guides. Cisco offers e-learning subscriptions for self-paced learning. Training partners offer instructor-led courses.

Q291 What is the role of a network engineer in an organisation?

Ans A network engineer designs, implements, and maintains the company's network infrastructure. Day-to-day tasks: configure new switches and routers, troubleshoot connectivity issues, implement security policies, plan for network growth, maintain documentation, coordinate with ISP for WAN services, respond to network outages, and implement changes through the change management process. Senior engineers also design new network architectures and evaluate new technologies.

Q292 What is a network diagram and why is it essential?

Ans A network diagram is a visual map of your network showing devices, connections, IP addresses, and VLANs. Types: Physical diagram (actual hardware locations, cable runs, rack layout). Logical diagram (IP addressing, VLANs, routing). Without accurate diagrams: troubleshooting takes hours instead of minutes. Onboarding new engineers takes weeks. During an outage, you cannot quickly identify the impact. Tools: Microsoft Visio (paid), draw.io (free), Lucidchart, NetBox. Keep diagrams updated after every change.

Q293 What is the change management process in networking?

Ans Formal change process: Submit change request (what, when, why, rollback plan). Risk assessment (what could go wrong?). Change Advisory Board (CAB) approval. Implementation window (usually nights/weekends). Implement change (one change at a time). Verify (test connectivity, check logs). Document results. Why? An uncontrolled network change causes half the outages. With proper change management: everyone knows what changed, rollback is planned, business impact is minimised.

Q294 What is a network audit?

Ans A network audit reviews all devices and configurations against security standards and best practices. Audit checklist: all default passwords changed, SSH enabled and Telnet disabled, no unused open ports, firmware/IOS up to date, access control policies match current staff roles, unused ports disabled, logging enabled and monitored, backup configs in version control, documentation current and accurate. Regular audits catch security gaps before attackers do.

Q295 What additional certifications complement CCNA?

Ans Recommended complementary certifications: CompTIA Security+ (cybersecurity fundamentals — pairs perfectly with CCNA for security roles), AWS Solutions Architect Associate (cloud networking), Microsoft AZ-900/AZ-104 (Azure networking), CompTIA Network+ (vendor-neutral networking breadth), Cisco Certified DevNet Associate (network automation), CCNP Enterprise (next level Cisco). For security focus: CCNA + Security+ + CCNP Security is a powerful combination.

Q296 What is the value of CCNA in the Indian job market?

Ans CCNA is highly valued in India's IT sector. Major employers: TCS, Infosys, Wipro, HCL (IT services for global clients), Airtel, Jio, BSNL (telecom infrastructure), Indian banking sector, government organisations (BSNL, NIC, DRDO), MNCs with India offices (Cisco, IBM, Accenture), and security firms. With 4+ years experience (like yours) plus CCNA, roles in network architecture and security design become accessible. Bihar government IT roles also require networking certifications.

TIP *Your 4+ years IT experience at Chandan Steel plus CCNA creates a strong combined profile!*

Q297 What is IPv6 adoption and why should network engineers learn it?

Ans IPv4 addresses are officially exhausted — all regional registries have run out. IPv6 is the solution with unlimited addresses. Current state: most internet traffic still IPv4 due to NAT. Major cloud providers (AWS, Azure) now primarily IPv6. Mobile networks (Jio, Airtel 4G/5G) increasingly IPv6. Government networks globally mandating IPv6. Network engineers who deeply understand IPv6 are increasingly valuable. CCNA includes IPv6 addressing and basic configuration.

Q298 How do network engineers stay current with technology?

Ans Staying current: Cisco learning Network and DevNet blogs (new technologies). YouTube channels: Jeremy's IT Lab, David Bombal, Network Chuck. Podcasts: Packet Pushers, Cisco Champion Radio. Twitter/LinkedIn: follow Cisco, Ivan Pepelnjak, Russ White. Cisco Live (annual conference — free virtual attendance). Home lab or GNS3 for testing new technologies. Study path: CCNA → CCNP → specialist certification → hands-on project experience. Never stop learning.

Q299 What is the difference between a network architect and a network engineer?

Ans Network Engineer: implements and maintains existing network designs. Configures devices. Troubleshoots issues. Day-to-day operational focus. Network Architect (typically 10+ years experience, CCIE level): designs entire network infrastructure from scratch. Makes strategic technology choices (which routing protocol, which hardware platform, cloud vs on-premises, SD-WAN design). Presents architecture to executives. Larger scope, strategic thinking, less hands-on CLI work. Both need strong fundamentals.

Q300 What final advice would you give to a CCNA student?

Ans From one network professional to another: Master subnetting until it is automatic — do it in your sleep. Understand the WHY behind every protocol — do not just memorise commands. Build labs, break them, fix them — that is how real learning happens. Connect with other students and engineers on forums (Cisco Learning Network, Reddit r/ccna). Your real-world IT experience (FortiGate, Active Directory, VPN, DHCP) is a massive advantage — connect CCNA concepts to what you already know. The CCNA is not the finish line — it is the starting line. Best of luck!

TIP *Akhilesh, your 4+ years in IT infrastructure is a tremendous foundation. CCNA will formalise and deepen that knowledge. You've got this!*